

Introduction to Computer Vision

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Today's agenda

- ✓ What are we going to learn in computer vision?
- ✓ Vision datasets
- ✓ Vision tasks
- ✓ Challenges in visual recognition
- ✓ History of the recognition tasks
- ✓ Course administrative details

MASSACHUSETTS INSTITUTE OF TECHNOLOGY
PROJECT MAC

Artificial Intelligence Group
Vision Memo. No. 100.

July 7, 1966

THE SUMMER VISION PROJECT

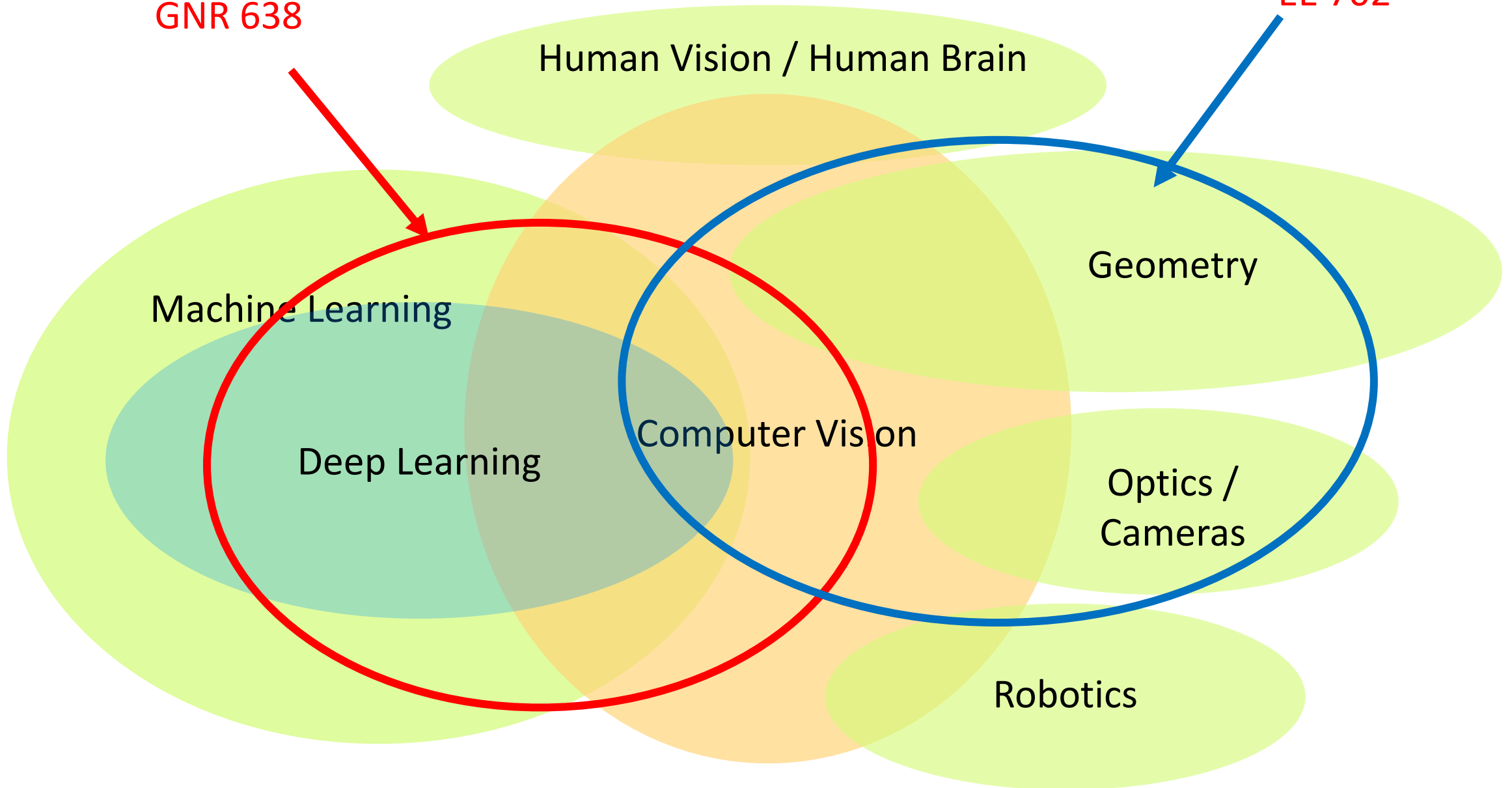
Seymour Papert

The summer vision project is an attempt to use our summer workers effectively in the construction of a significant part of a visual system. The particular task was chosen partly because it can be segmented into sub-problems which will allow individuals to work independently and yet participate in the construction of a system complex enough to be a real landmark in the development of "pattern recognition".

<https://dspace.mit.edu/handle/1721.1/6125>

GNR 638

EE 702



Computer Vision

Make computers understand images and video.



What kind of scene?

Where are the cars?

How far is the building?

...

Requires multi-level processing, integrating visual and semantic cues

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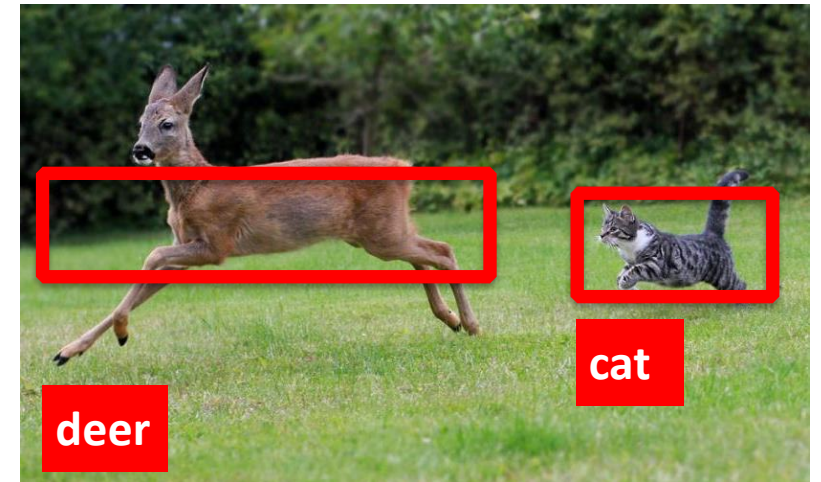
Relationship with Other Fields

- Image Processing: Image $\longrightarrow$ Image



Relationship with Other Fields

- Computer Vision: Image $\longrightarrow$ Knowledge

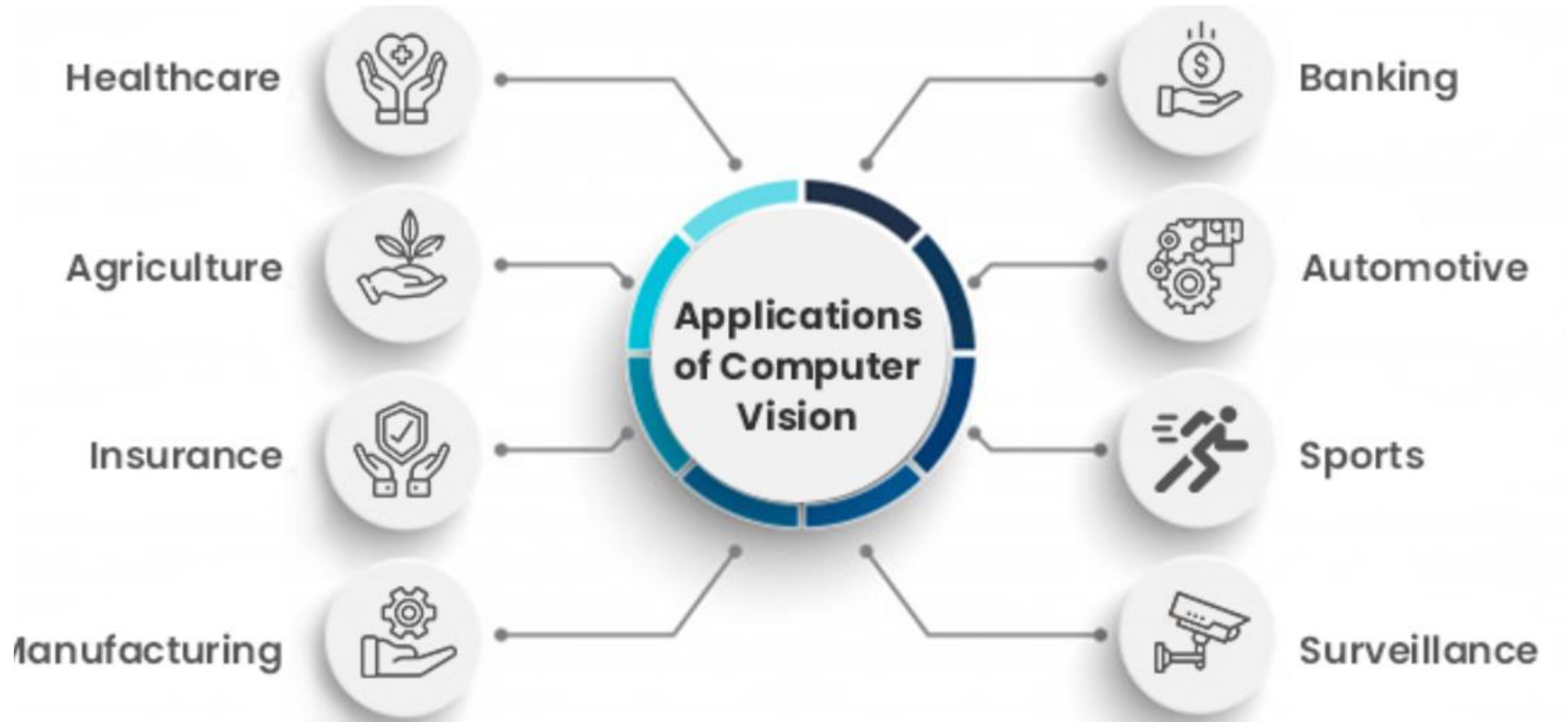


Relationship with Other Fields

- Computer Graphics: Knowledge $\longrightarrow$ Image

Vertices, Locations, Objects,
Shapes, Colors, Material properties,
Lighting settings, Camera settings, etc.





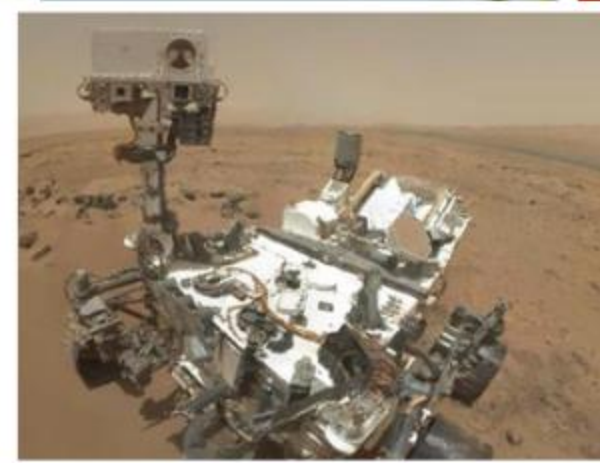


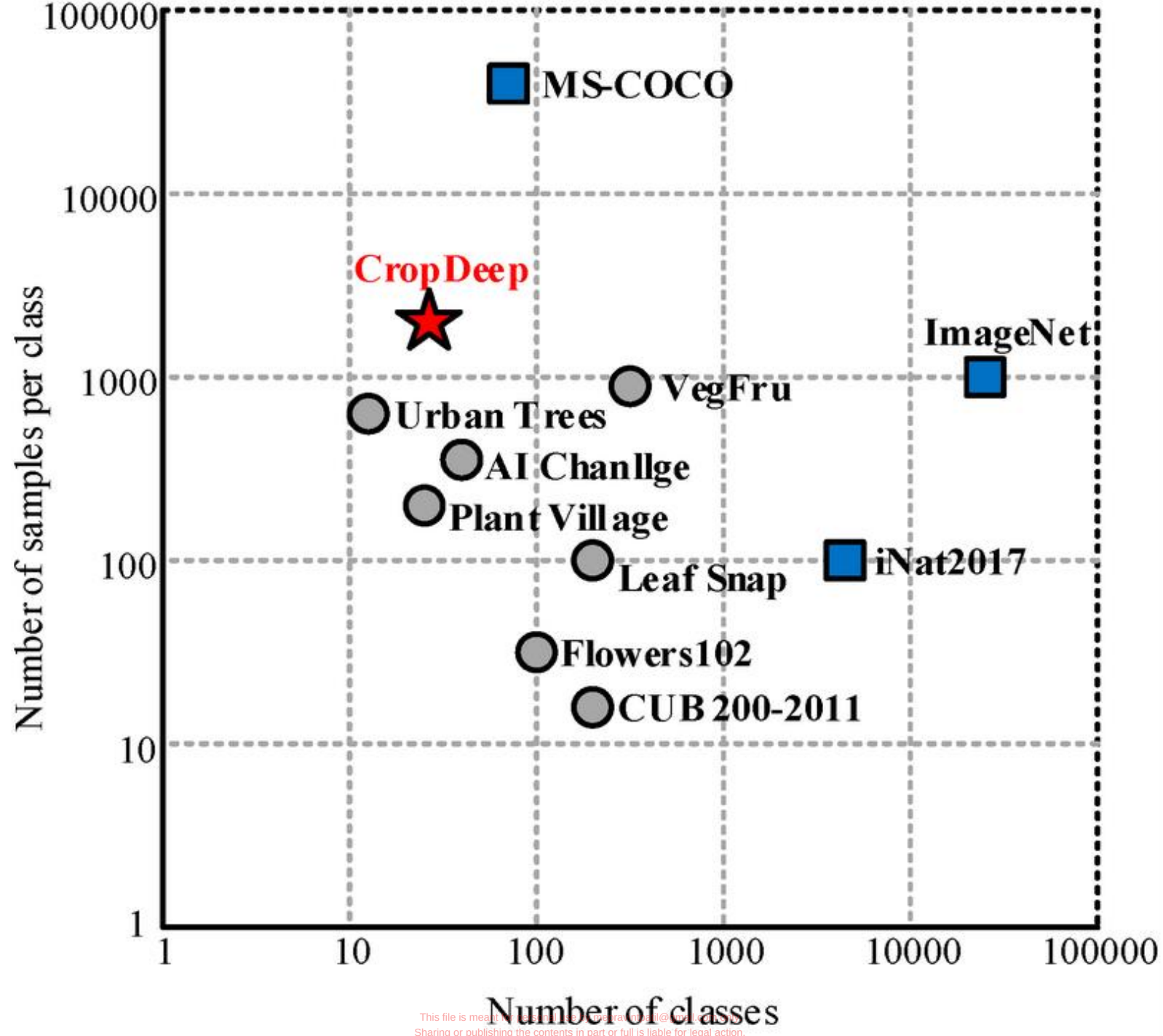
The European Camera of the Year 1983.

Fortunately, fine photography is a universal language.

"The income between the total technological productivity of the economy and the aggregate demand" - John Maynard Keynes's *Principles of Economics*. Great Britain.

They show the Pentax cameras most likely to satisfy the photo enthusiast. The Super Program (135/35mm SLR), A, 135/35mm SLR, and Super Speed (135/35mm SLR) are all available in 135/35mm SLR format. The Super Program is the most popular, with a top speed of 1/2000, 450-1000, 135/35mm SLR, and 135/35mm SLR. The Super Speed is the most popular, with a top speed of 1/2000, 450-1000, 135/35mm SLR, and 135/35mm SLR. The Super Program is the most popular, with a top speed of 1/2000, 450-1000, 135/35mm SLR, and 135/35mm SLR. The Super Speed is the most popular, with a top speed of 1/2000, 450-1000, 135/35mm SLR, and 135/35mm SLR.





ImageNet dataset (wiki)



ImageNet-21K [\[edit\]](#)

The full original dataset is referred to as ImageNet-21K. ImageNet-21k contains 14,197,122 images divided into 21,841 classes. Some papers round this up and name it ImageNet-22k.^[29]

The full ImageNet-21k was released in Fall of 2011, as `fall11_whole.tar`. There is no official train-validation-test split for ImageNet-21k. Some classes contain only 1-10 samples, while others contain thousands.^[29]

ImageNet-1K [\[edit\]](#)

There are various subsets of the ImageNet dataset used in various context, sometimes referred to as "versions".^[16]

One of the most highly used subset of ImageNet is the "ImageNet Large Scale Visual Recognition Challenge (ILSVRC) 2012–2017 image classification and localization dataset". This is also referred to in the research literature as ImageNet-1K or ILSVRC2017, reflecting the original ILSVRC challenge that involved 1,000 classes. ImageNet-1K contains 1,281,167 training images, 50,000 validation images and 100,000 test images.^[30]

Each category in ImageNet-1K is a leaf category, meaning that there are no child nodes below it, unlike ImageNet-21K. For example, in ImageNet-21K, there are some images categorized as simply "mammal", whereas in ImageNet-1K, there are only images categorized as things like "German shepherd", since there are no child-words below "German shepherd".^[22]

LAION-400-MILLION OPEN DATASET

by: Christoph Schuhmann, 20 Aug, 2021

We present LAION-400M: 400M English (image, text) pairs - see also our [Data Centric AI NeurIPS Workshop 2021 paper](#)

🔗 Concept and Content

The LAION-400M dataset is entirely openly, freely accessible.

WARNING: be aware that this large-scale dataset is non-curated. It was built for research purposes to enable testing model training on larger scale for broad researcher and other interested communities, and is **not** meant for any real-world production or application.

We have filtered all images and texts in the LAION-400M dataset with OpenAI's [CLIP](#) by calculating the cosine similarity between the text and image embeddings and dropping those with a similarity below 0.3. The threshold of 0.3 had been determined through human evaluations and seemed to be a good heuristic for estimating semantic image-text-content matching.

The image-text-pairs have been extracted from the [Common Crawl](#) web data dump and are from random web pages crawled between 2014 and 2021.

blue cat



Russian Blue Postcards (Package of 8)



CAT, LED Lamps, slingly, FamilyTrophy.com - Family...



Halloween, Cat, Paper Plate, Toddler, Preschool



Blue Wooden Cat



Ink sketch of a cat



Blue Eyes Greeting Card



Alicia Vannoy Call Framed Prints - Cat - Kitten Bl...



Blues for a Black Cat and Other Stories



Sitting cat low poly in Gloss Blue Porcelain



Tonkinese cat drawing BZTAT



Blue Cat Full Moon Postcard



Kitten Digital Art - Soft Kitty by William Paul M...



Blue and Black Panther Football Colors Acrylic Pai...



Tonkinese drawing - picture #10



Cat with blue eyes



Pretty Blue Cat cards

cat with blue eyes



cats with different colored through golden this ca...



cat, white, and eyes image



Siamese cat lying in the bast basket



Catito, Blue Eyes by RBenedetti



Pam Pam, le Chat Blanc aux Yeux de 2 Couleurs qu...



heterochromia-cat-cross-eyed-alos-5



iPhone Wallpaper Two white kittens, blue eyes, pla...



5D Diamond Painting White Cat with Blue Eyes Kit



Coby the Cat with Piercing Blue Eyes, over 1 Milli...



heterochromia-cat-cross-eyed-alos-17



5D Diamond Painting White Cat with Blue Eyes Kit



ragdoll stock photo © nailaschwarz



This Cat Has The Most Stunningly Beautiful Blue Ey...



A cat conquers the Internet with blue eyes



cat, white, and eyes image



blue eyes and cute kitty image

Video recognition - ActivityNet

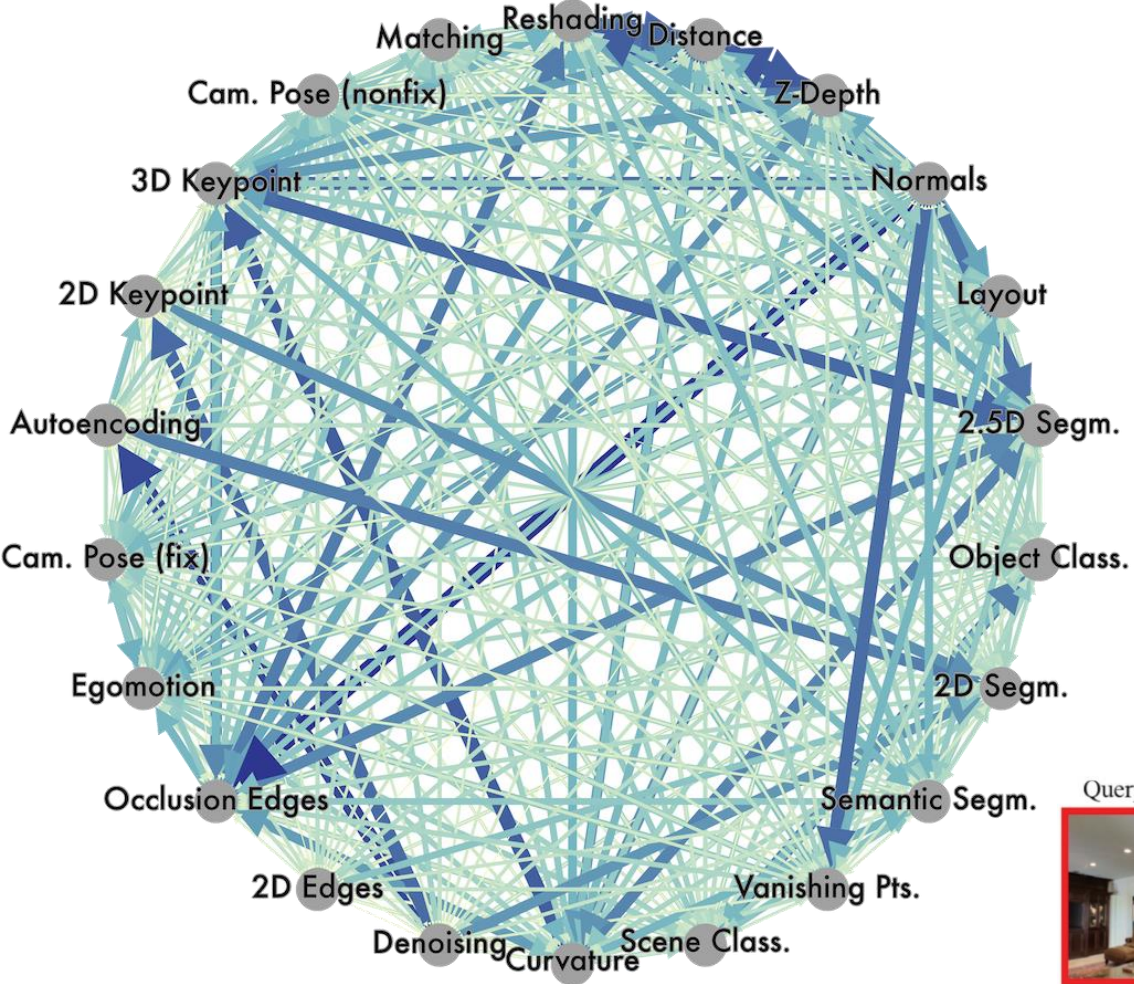


Figure 1. *ActivityNet* organizes a large number of diverse videos that contain human activities into a semantic taxonomy. **Top-row** shows the root-leaf path for the activity *Cleaning windows*. **Bottom-row** shows the root-leaf path for the activity *Brushing teeth*. Each box illustrates example videos that lie within the corresponding taxonomy node. Green intervals indicate the temporal extent of the activity. All figures are best viewed in color.

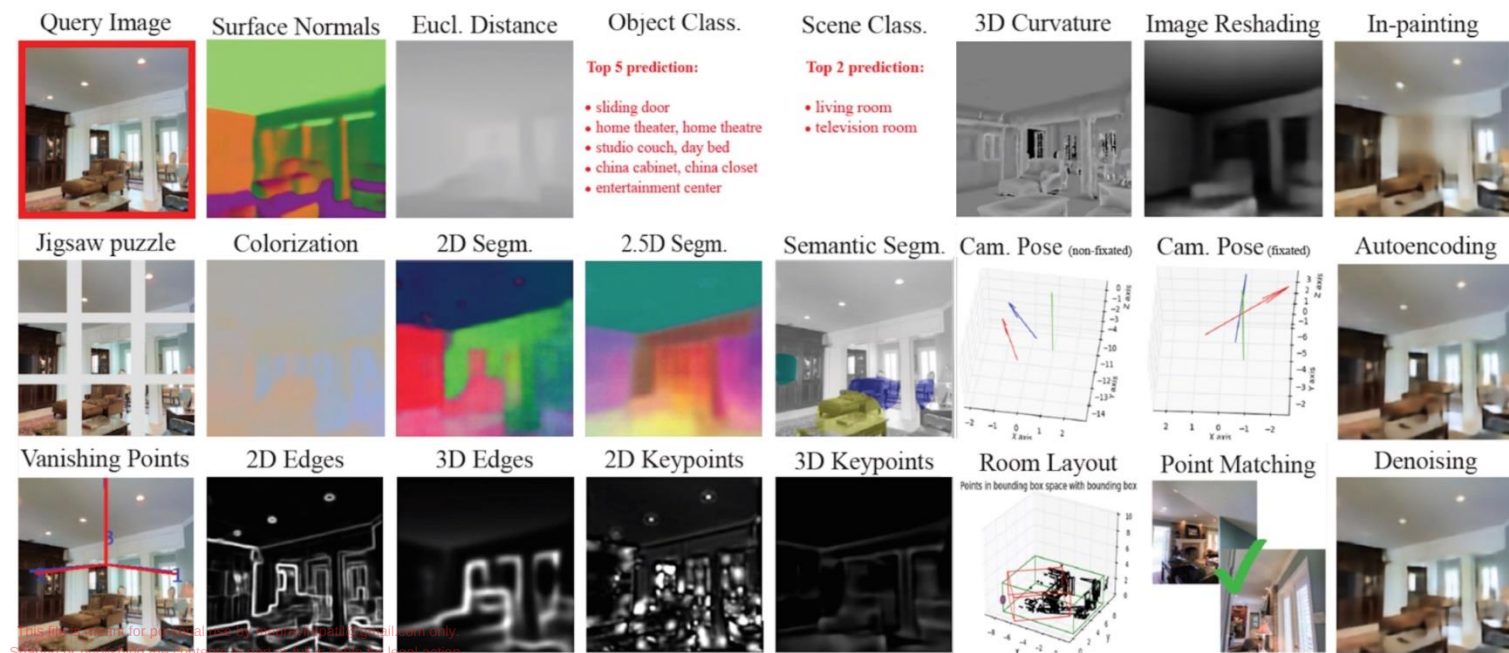
The daily action dataset

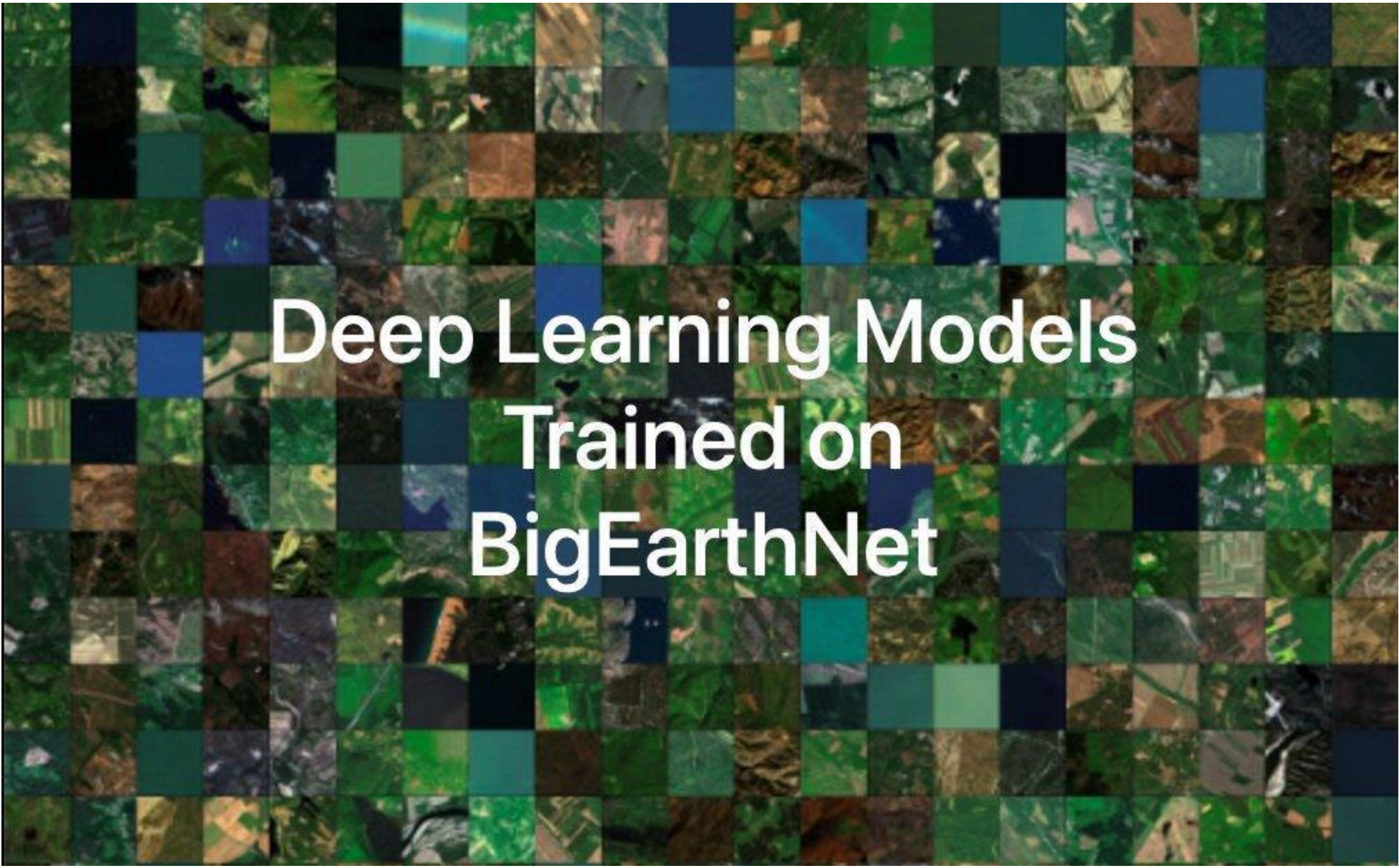


Figure 3. Frame and annotation samples of CDAD, including multi-person action (e.g. shake hands, hug), single person action (e.g. raise hand), human-object interaction action (e.g. smoke, sweep). Spatial-temporal annotations are provided for all actions. “Green” indicates temporal annotation of action instances, “blue” represents background (no action), and red rectangles are spatial annotations.



Taskonomy





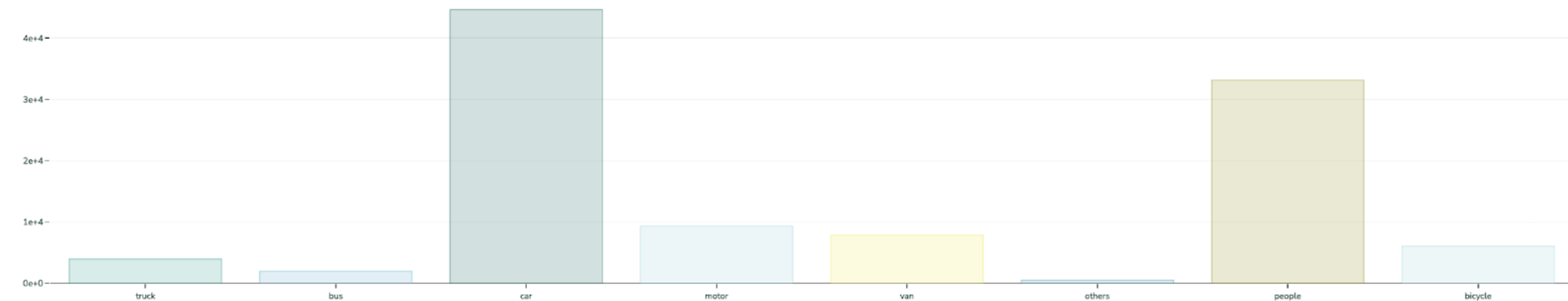
Deep Learning Models Trained on BigEarthNet

The issue of data imbalance is prominent

Assets 2000 - Objects 107368

Data Distribution

Click on any bar of the chart to search corresponding assets



What are the tasks we care about?

- What are the vision problems that can best be solved in a generalized manner using ML/DL techniques?

Verification

Is this a car?



Classification:

Is there a car in this picture?



Detection:

Where is the car in this picture?



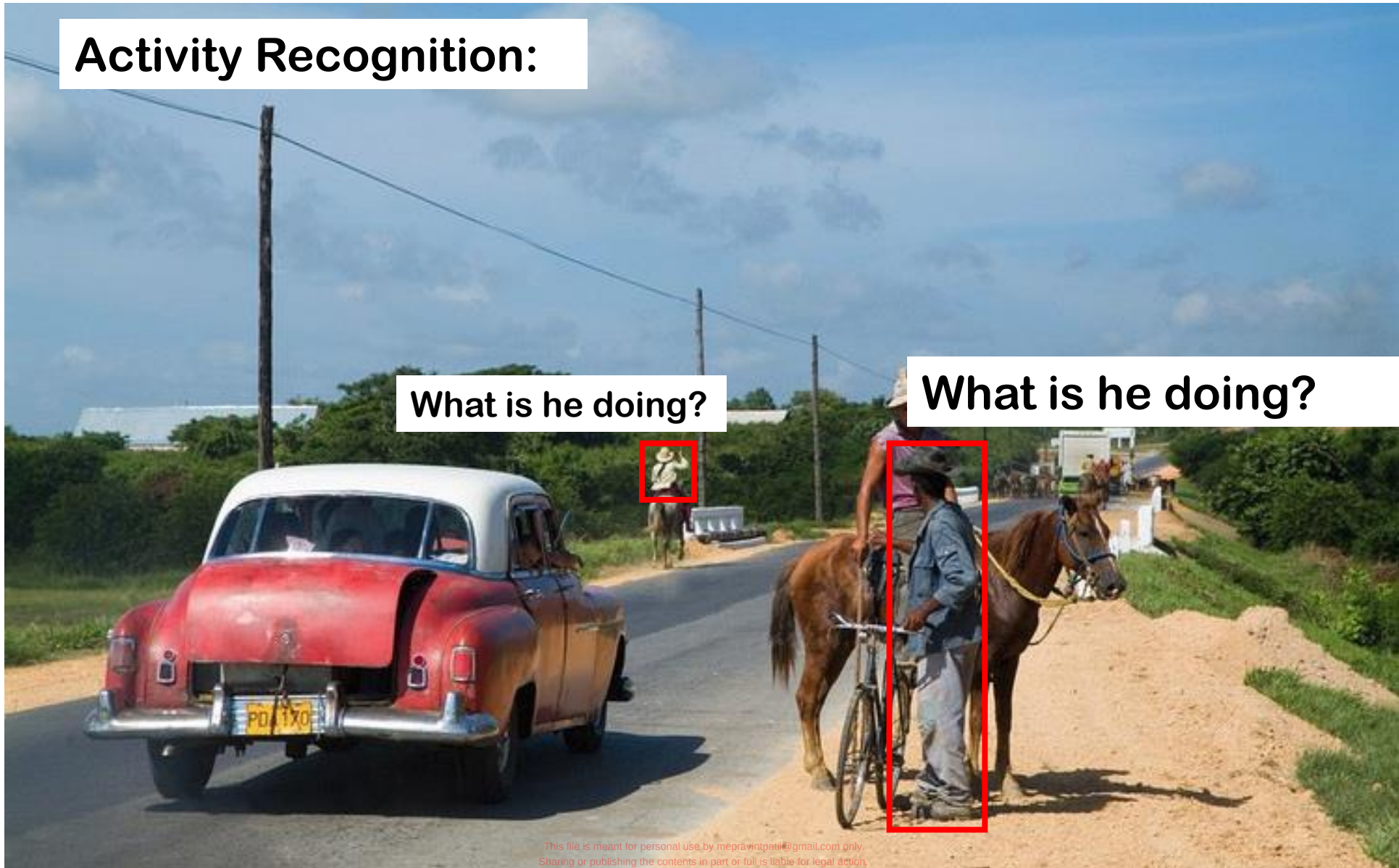
Pose Estimation:



Activity Recognition:

What is he doing?

What is he doing?



Object Categorization:

Sky

Person

Tree

Car

Horse

Person

Bicycle

Road

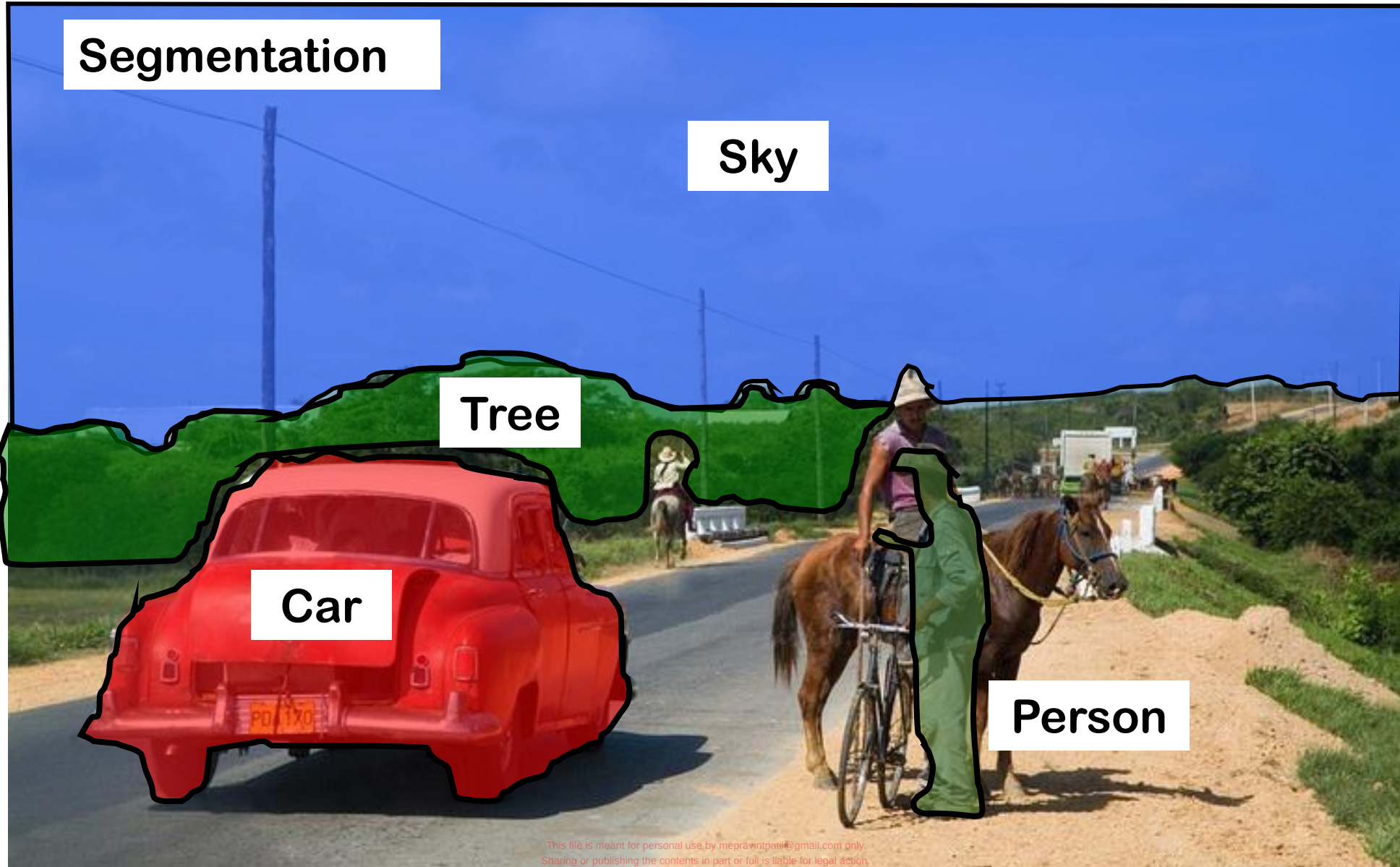
Segmentation

Sky

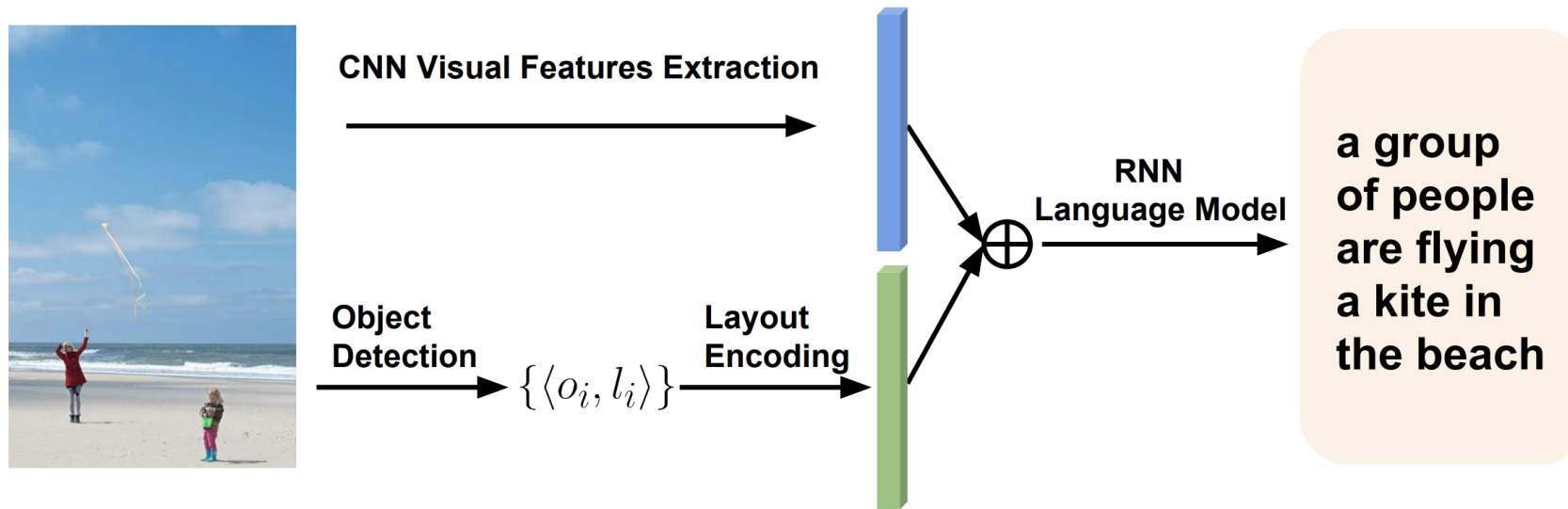
Tree

Car

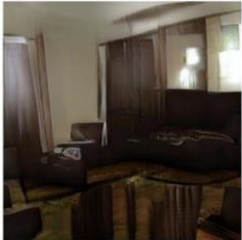
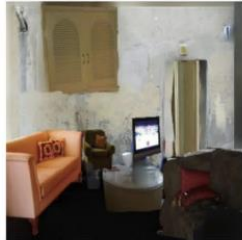
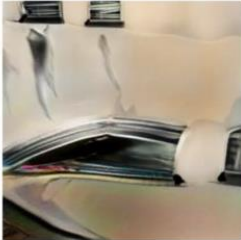
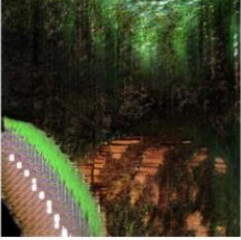
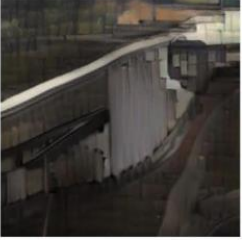
Person



Describing Images with Language



Text-to-Image Synthesis: Text2Scene

Input Caption	Real Image	SG2IM	HDGAN	AttnGAN	Text2Scene [no inpainting]	Text2Scene
A room with a <i>TV</i> and some different types of <i>couches</i> .						
A tall <i>monitor</i> is near a <i>keyboard</i> and a <i>mouse</i> .						
a <i>car bridge</i> going <i>over</i> a commuter <i>train</i> .						



A teddy bear swimming at the Olympics 400m Butterfly event.



A cute corgi lives in a house made out of sushi.



A cute sloth holding a small treasure chest. A bright golden glow is coming from the chest.



A brain riding a rocketship heading towards the moon.

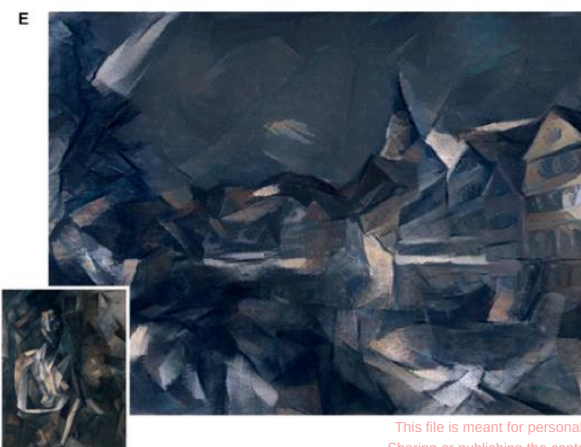


A dragon fruit wearing karate belt in the snow.



A strawberry mug filled with white sesame seeds. The mug is floating in a dark chocolate sea.

Neural style transfer



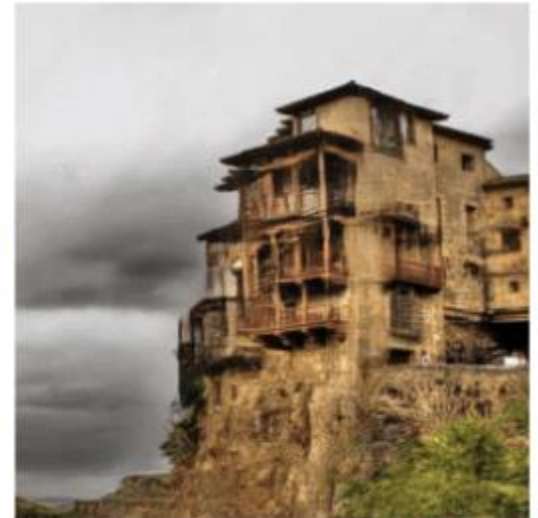
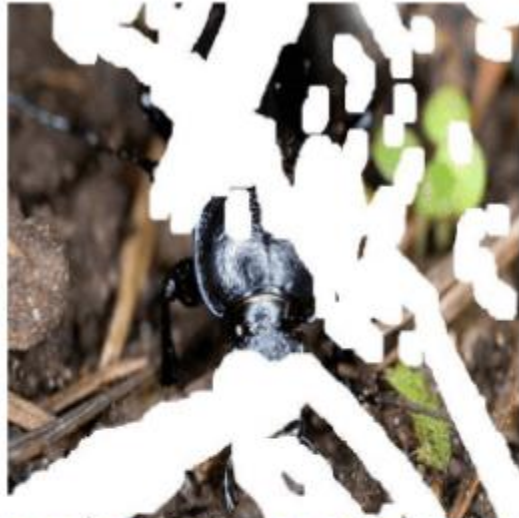


Image inpainting

Zebras ⇄ Horses



zebra → horse



horse → zebra

However, our main tasks to consider at the beginning of the course

Is this a dog?



Image Classification

What is there in image
and where?



Object Detection

Which pixels belong to
which object?

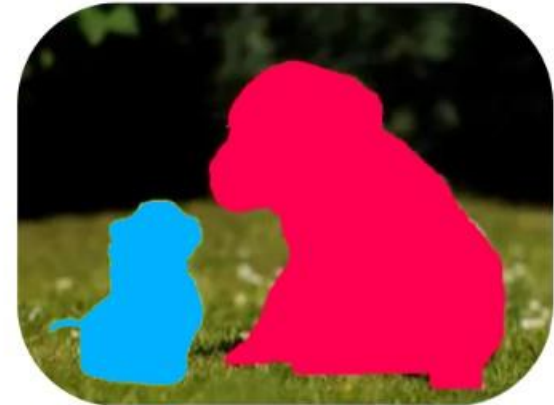


Image Segmentation

Why is recognition a difficult problem?

Challenges 1: view point variation



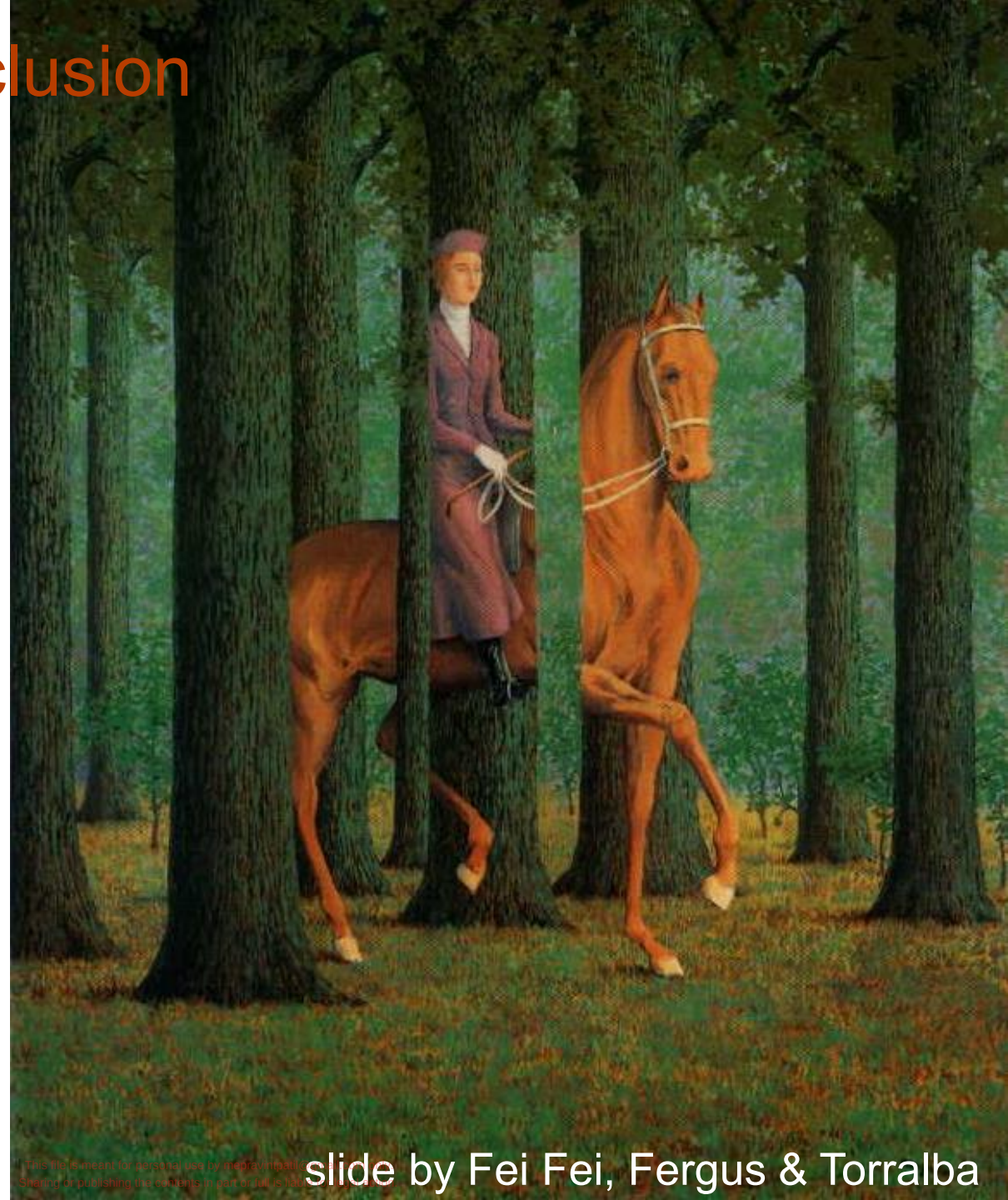
Michelangelo 1475-1564

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Challenges 2: illumination



Challenges 3: occlusion



Magritte, 1957

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slide by Fei Fei, Fergus & Torralba

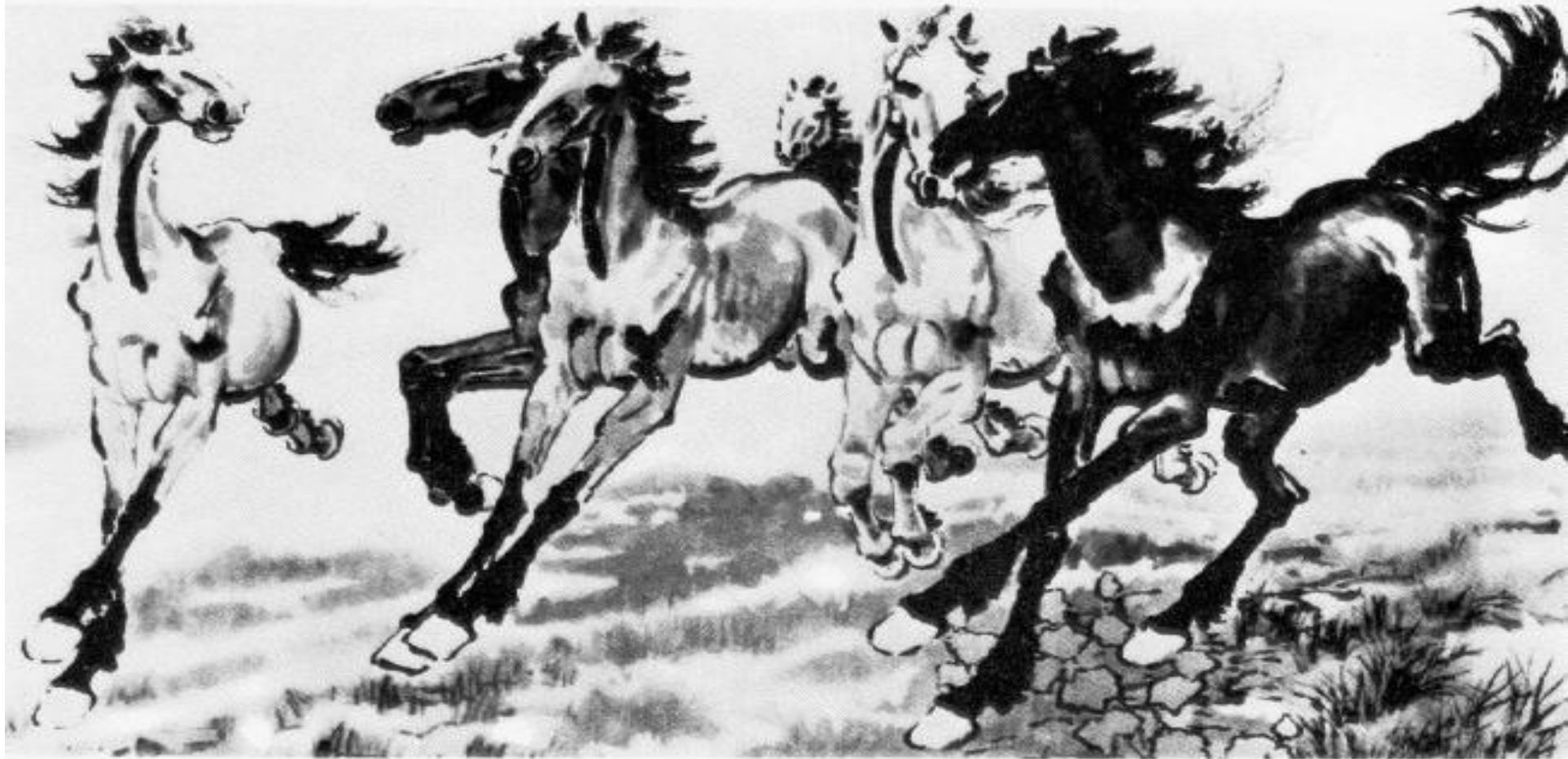
Challenges 4: scale



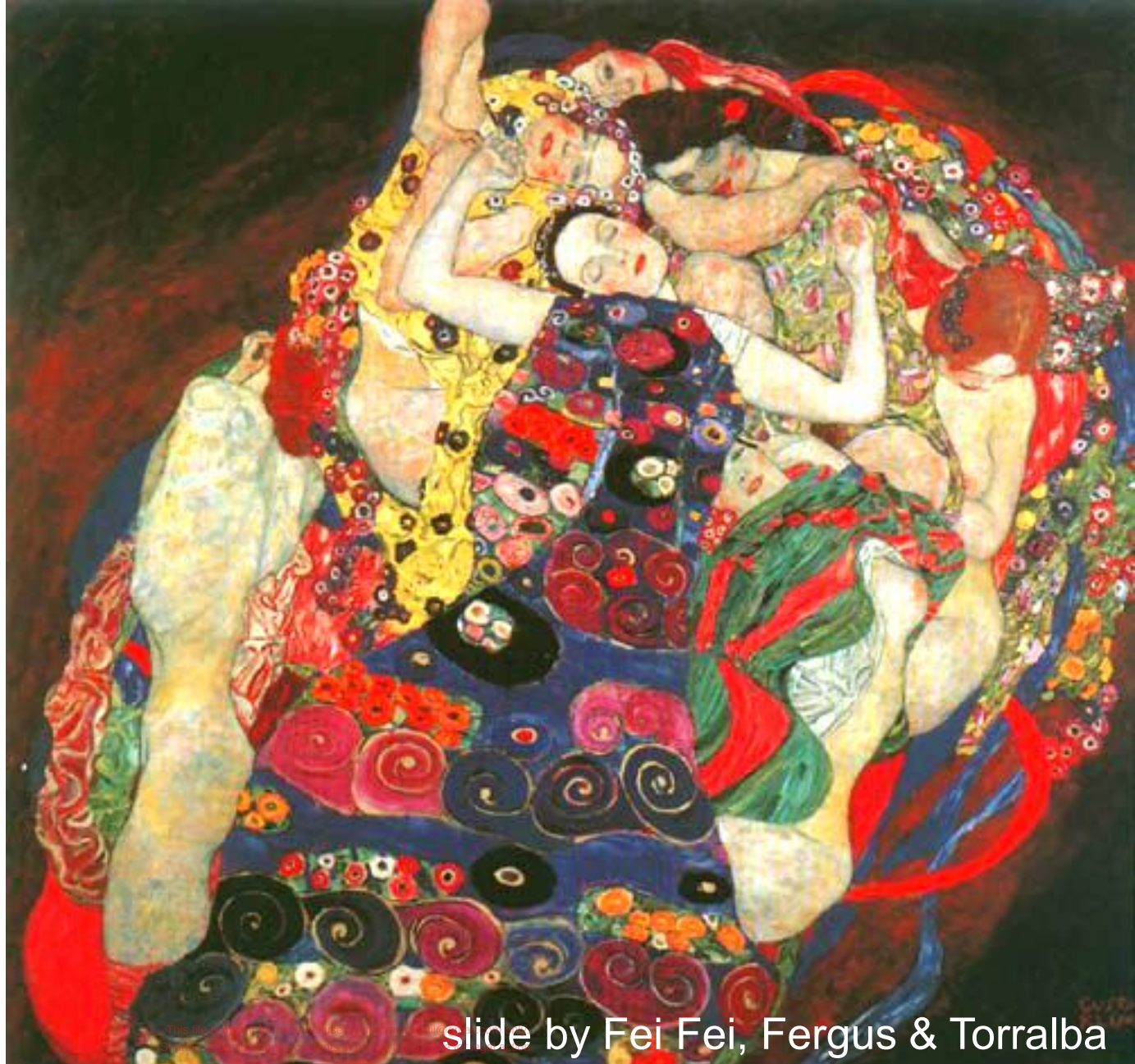
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slide by Fei Fei, Fergus & Torralba

Challenges 5: deformation



Challenges 6: background clutter



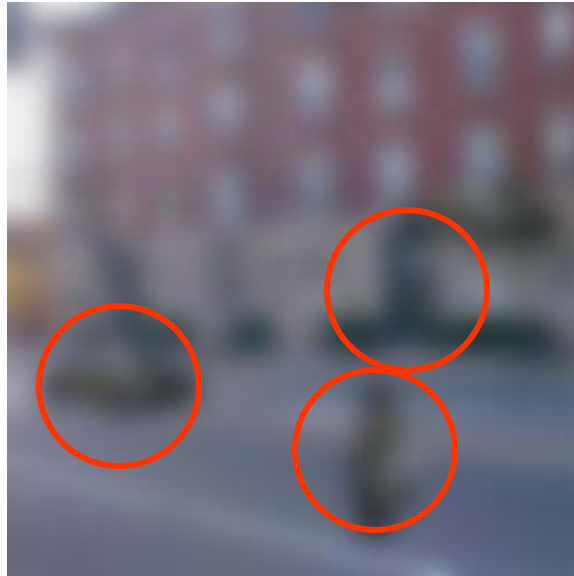
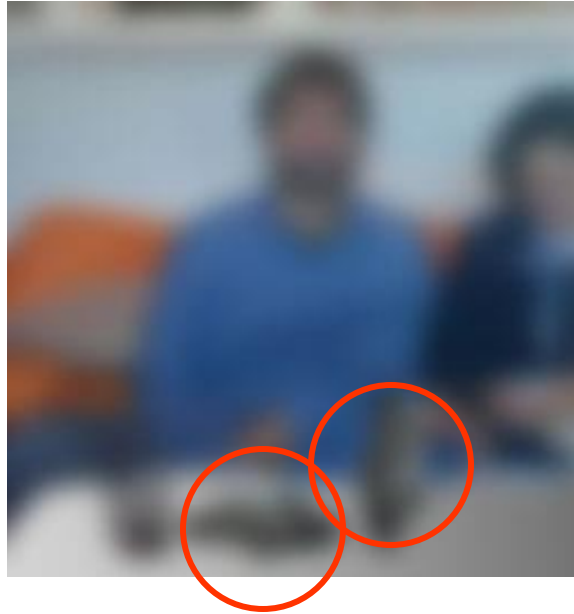
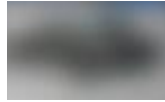
Klimt, 1913

slide by Fei Fei, Fergus & Torralba

Challenges 7: object intra-class variation

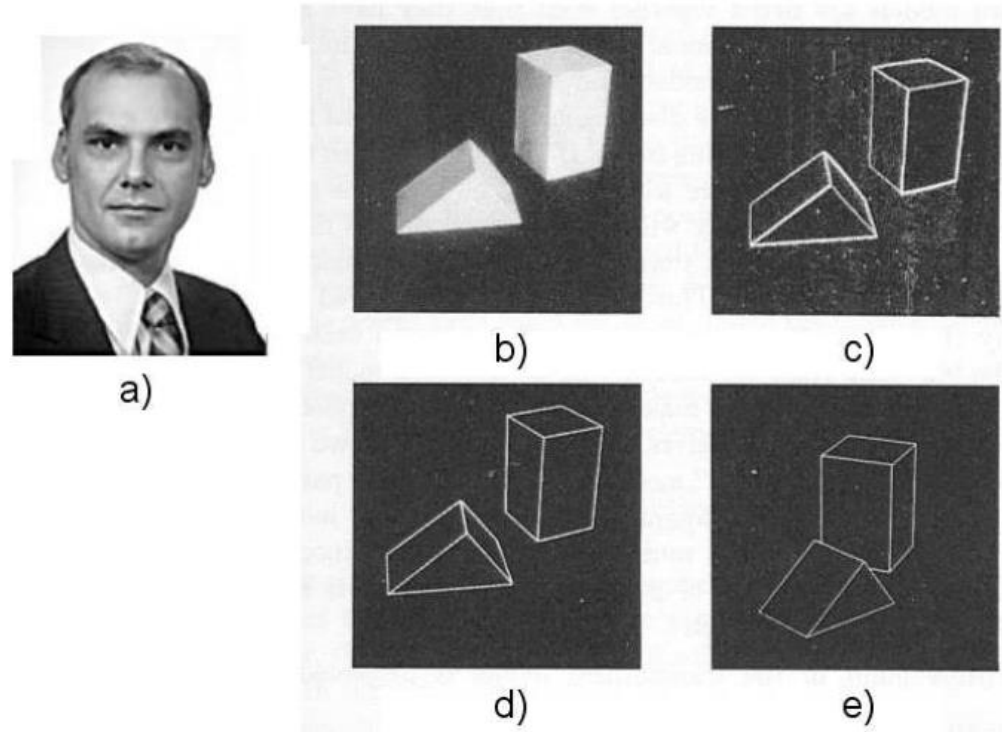


Challenges 8: local ambiguity



What are the progresses so far?

Recognition as an alignment problem: Block world

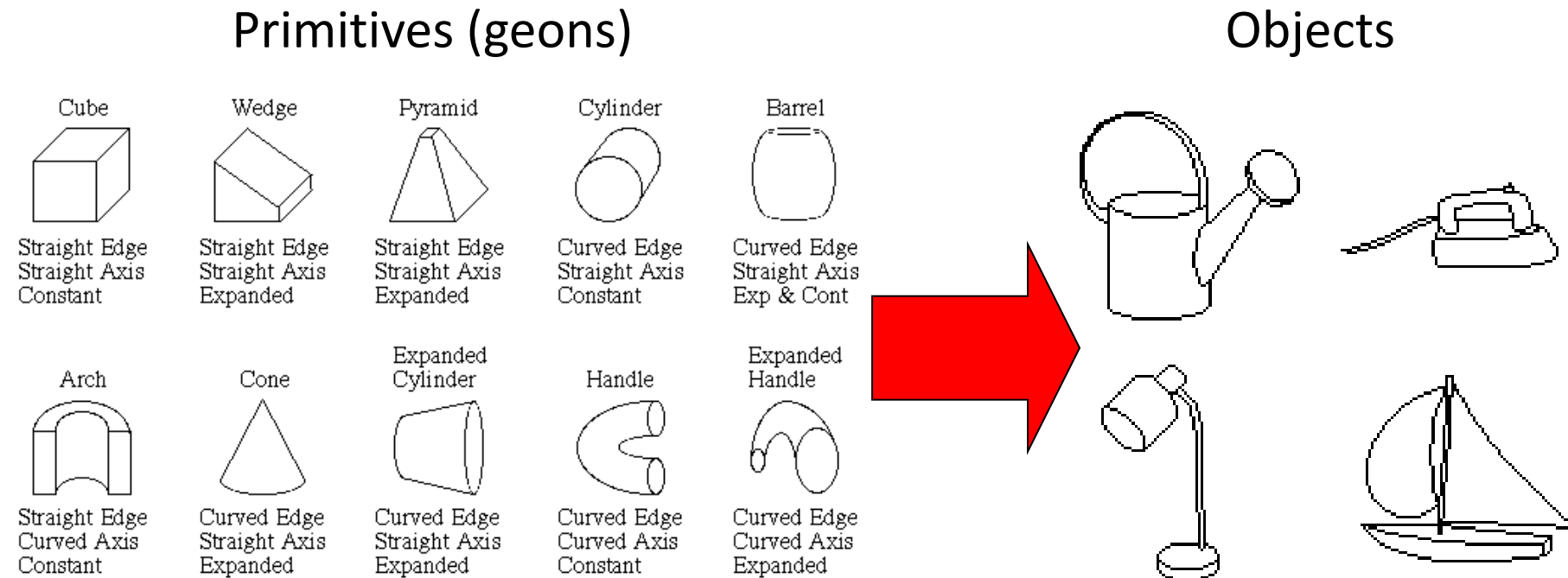


L. G. Roberts, [*Machine Perception of Three Dimensional Solids*](#), Ph.D. thesis, MIT Department of Electrical Engineering, 1963.

Fig. 1. A system for recognizing 3-d polyhedral scenes. a) L.G. Roberts. b) A blocks world scene. c) Detected edges using a 2x2 gradient operator. d) A 3-d polyhedral description of the scene, formed automatically from the single image. e) The 3-d scene displayed with a viewpoint different from the original image to demonstrate its accuracy and completeness. (b) - e) are taken from [64] with permission MIT Press.)

Recognition by components

Biederman (1987)



http://en.wikipedia.org/wiki/Recognition_by_Components_Theory

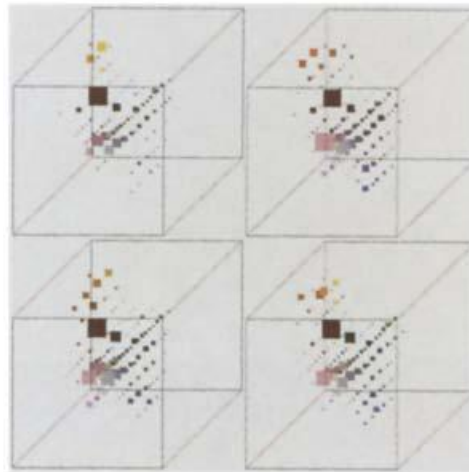
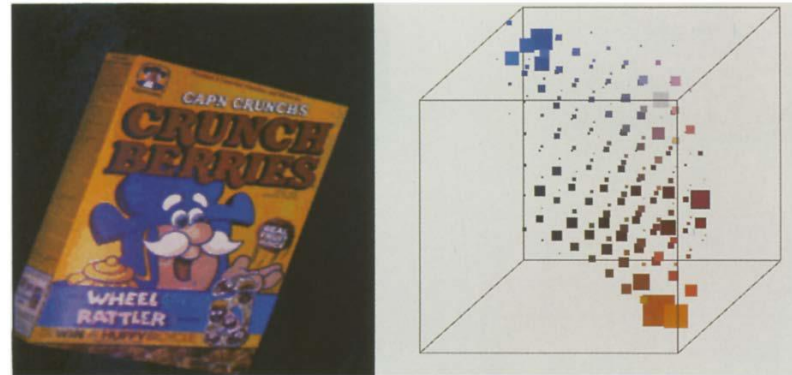
History of ideas in recognition

- 1960s – early 1990s: the geometric era
- **1990s: appearance-based models**
- Mid-1990s: sliding window approaches
- Late 1990s: local features
- Early 2000s: parts-and-shape models
- Mid-2000s: bags of features
- Present trends: combination of local and global methods, data-driven methods, context

Eigenfaces (Turk & Pentland, 1991)



Color Histograms



Swain and Ballard, [Color Indexing](#), IJCV 1991.

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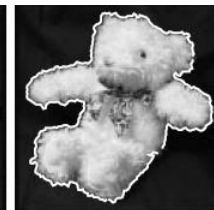
Sliding window approaches



History of ideas in recognition

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Local features for object instance recognition



D. Lowe (1999, 2004)

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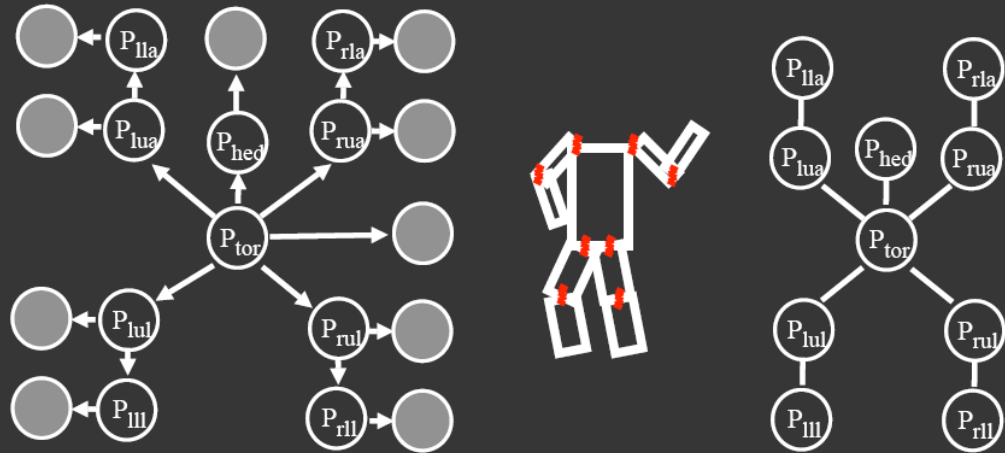
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Representing people

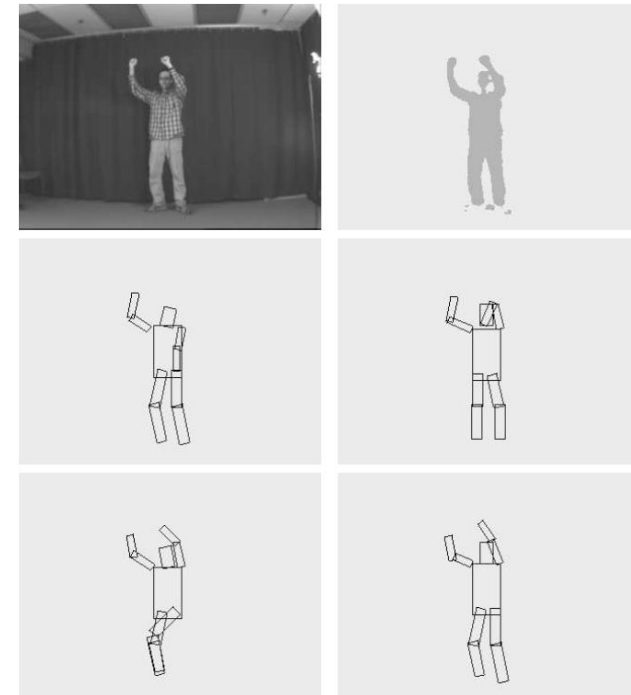
Pictorial structure model

Fischler and Elschlager(73), Felzenszwalb and Huttenlocher(00)



$$\Pr(P_{\text{tor}}, P_{\text{arm}}, \dots | \text{Im}) \propto \prod_{i,j} \Pr(P_i | P_j) \prod_i \Pr(\text{Im}(P_i))$$

↑
part geometry
↑
part appearance



History of ideas in recognition

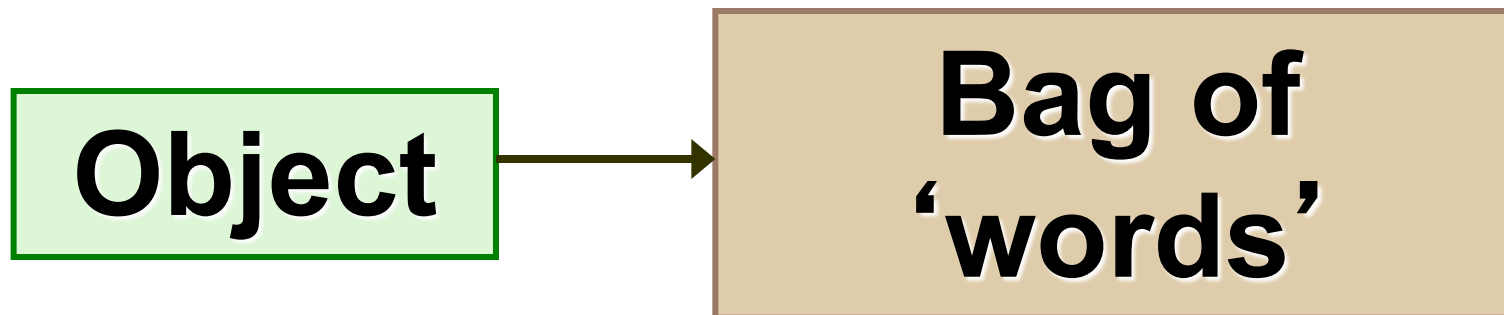
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Bag of words in NLP

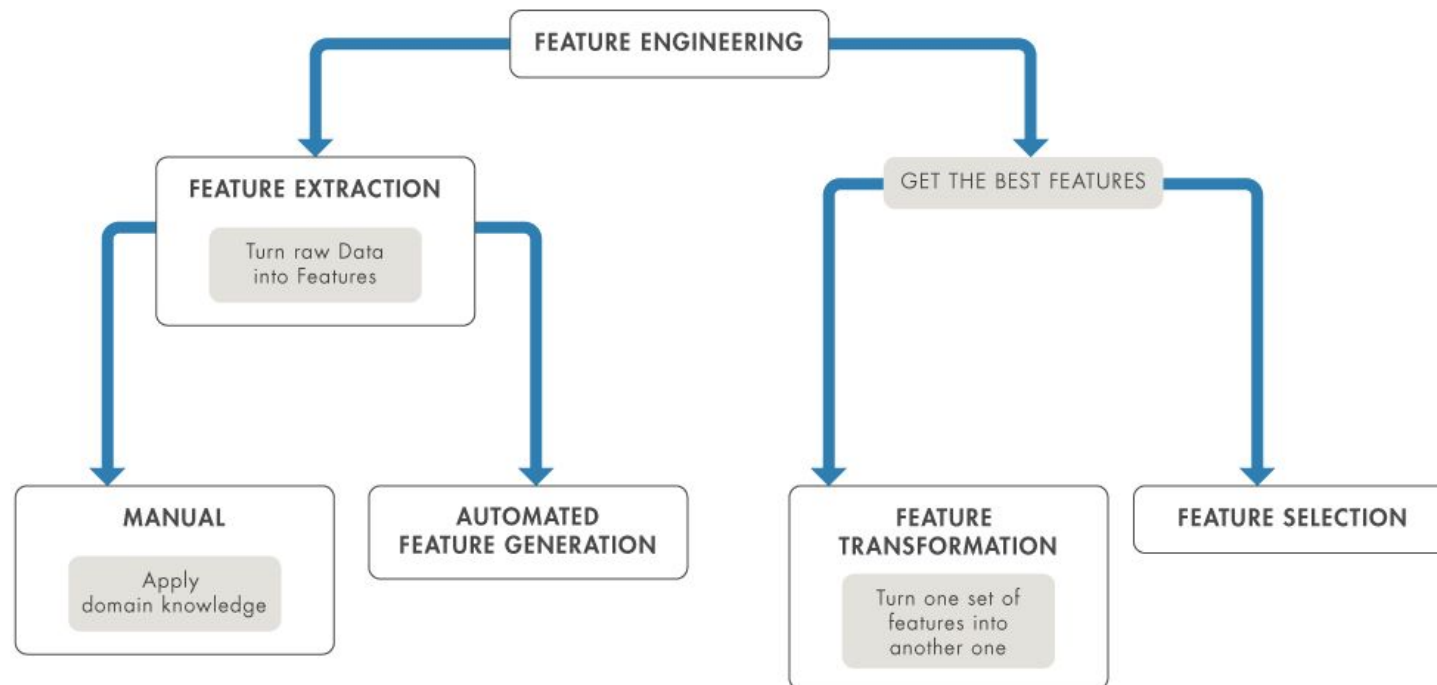


For each theme of paragraph, the bag contains theme-specific words

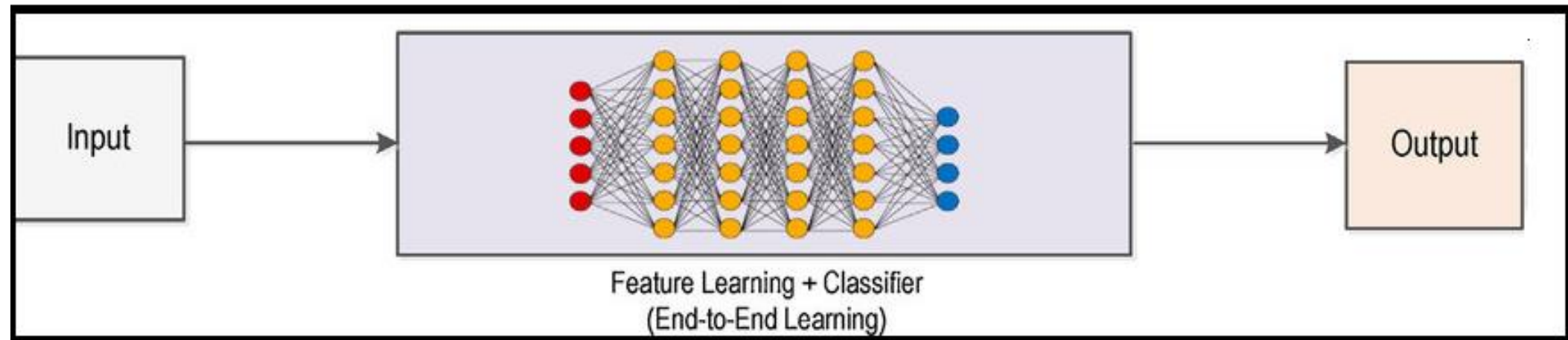
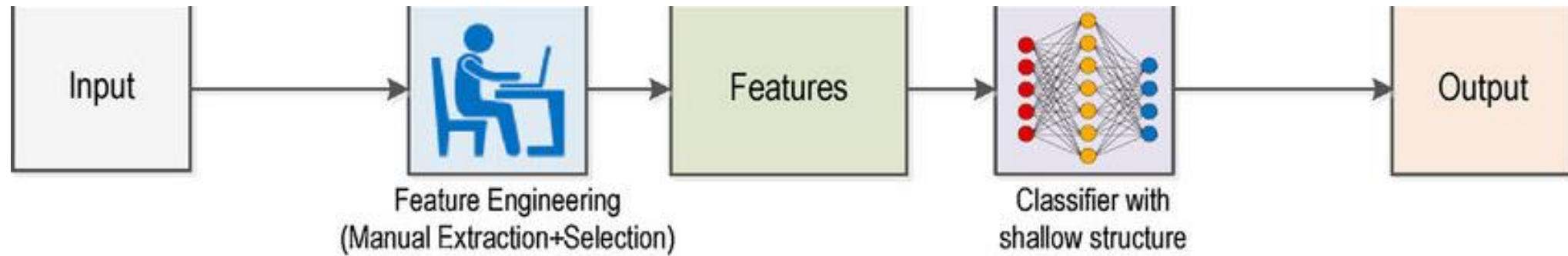
Can we do the same with images? What are visual words?



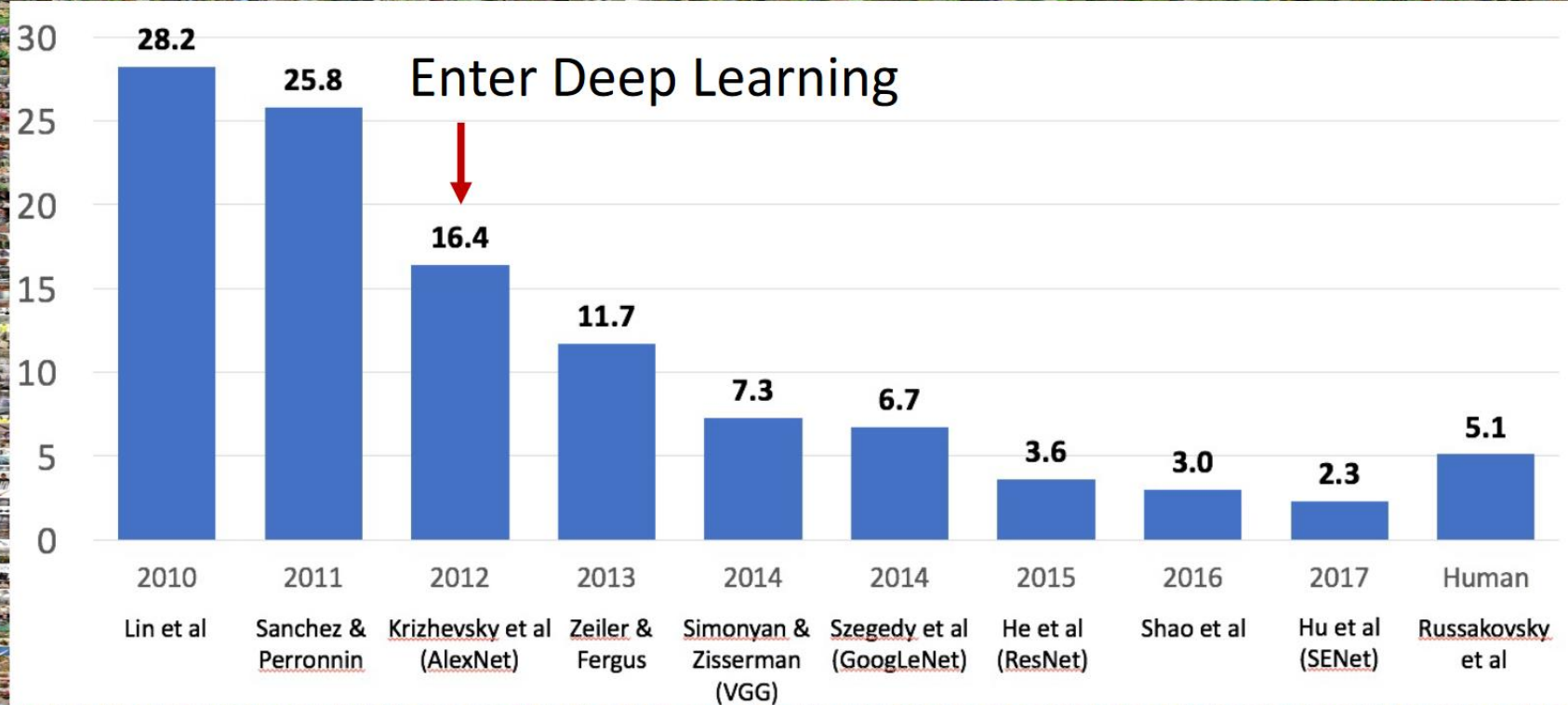
- ✓ All these techniques are based on manually defining features from the images
- ✓ Features – fixed length representations of the images/objects which are discriminative
- ✓ Manual feature extractions are ambiguous – which features are to be considered or discarded are not clear
- ✓ Many ad-hoc techniques follow to refine the features



The main idea of doing deep learning



IMAGENET Large Scale Visual Recognition Challenge



Perceptron

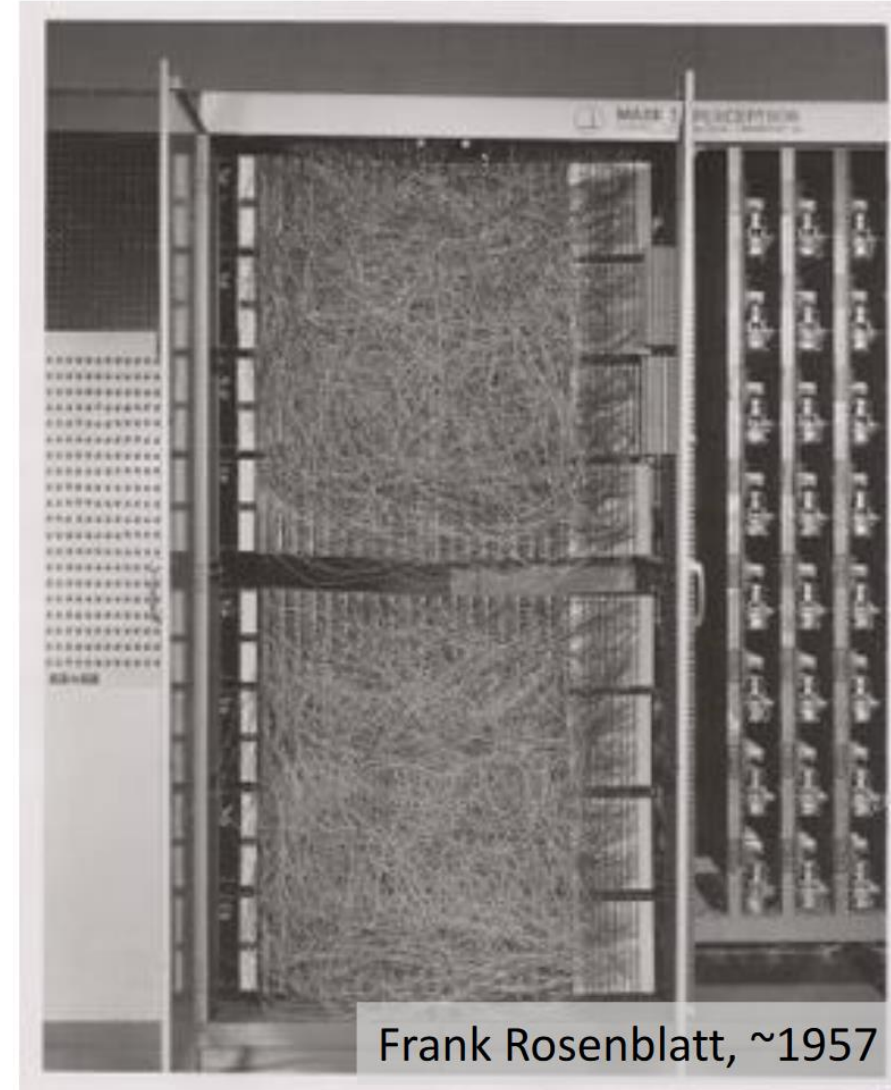
One of the earliest algorithms that could learn from data

Implemented in hardware! Weights stored in potentiometers, updated with electric motors during learning

Connected to a camera that used 20x20 cadmium sulfide photocells to make a 400-pixel image

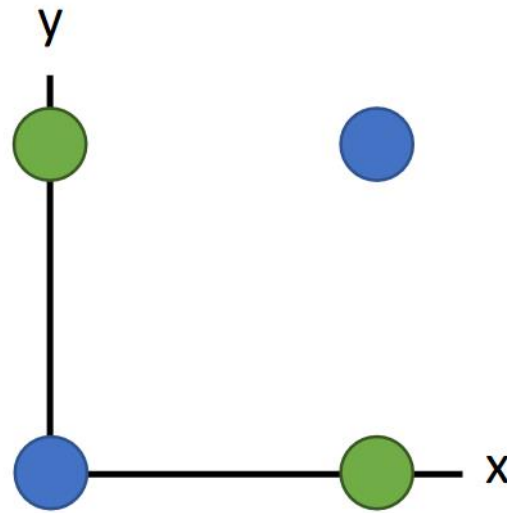
Could learn to recognize letters of the alphabet

Today we would recognize it as a **linear classifier**

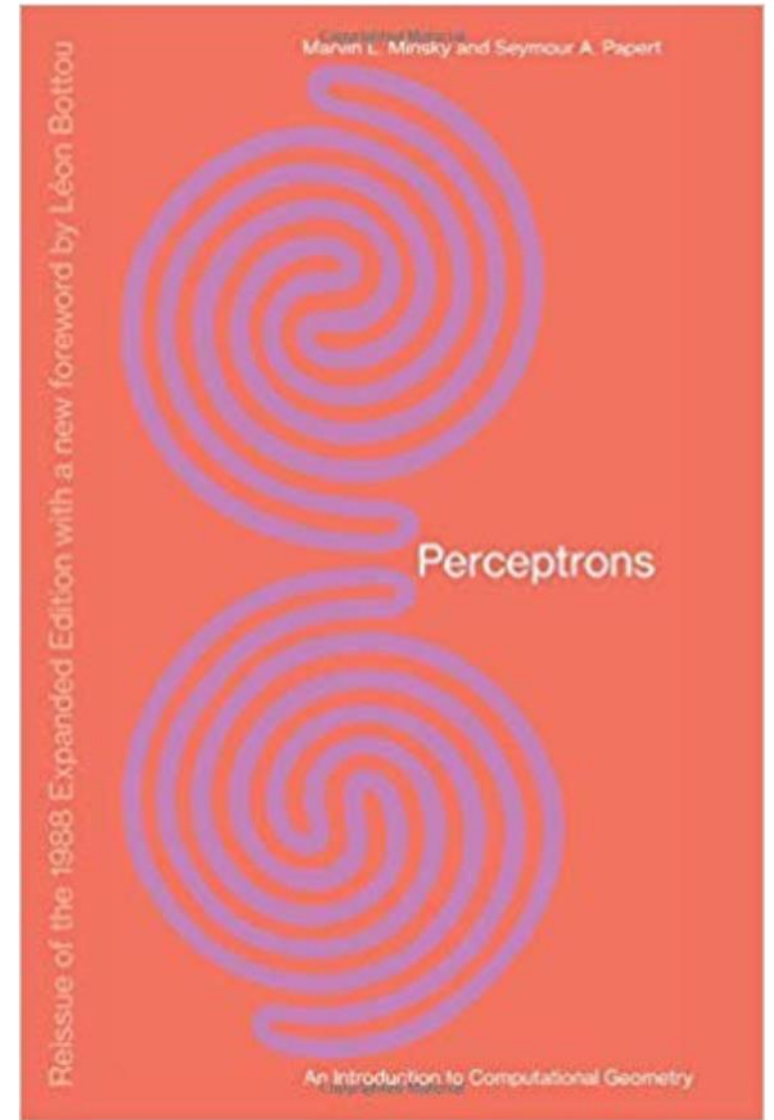


Minsky and Papert, 1969

X	Y	F(x,y)
0	0	0
0	1	1
1	0	1
1	1	0



Showed that Perceptrons could not learn the XOR function
Caused a lot of disillusionment in the field

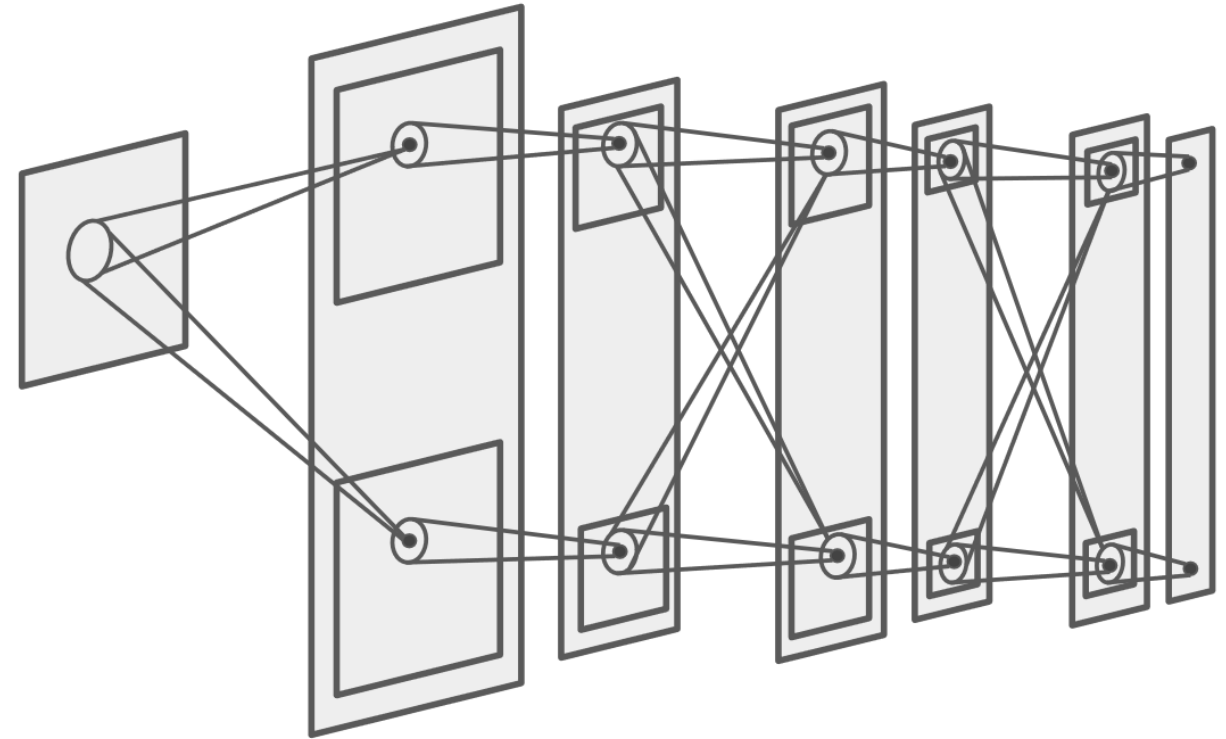


Neocognitron: Fukushima, 1980

Computational model the visual system,
directly inspired by Hubel and Wiesel's
hierarchy of complex and simple cells

Interleaved simple cells (convolution)
and complex cells (pooling)

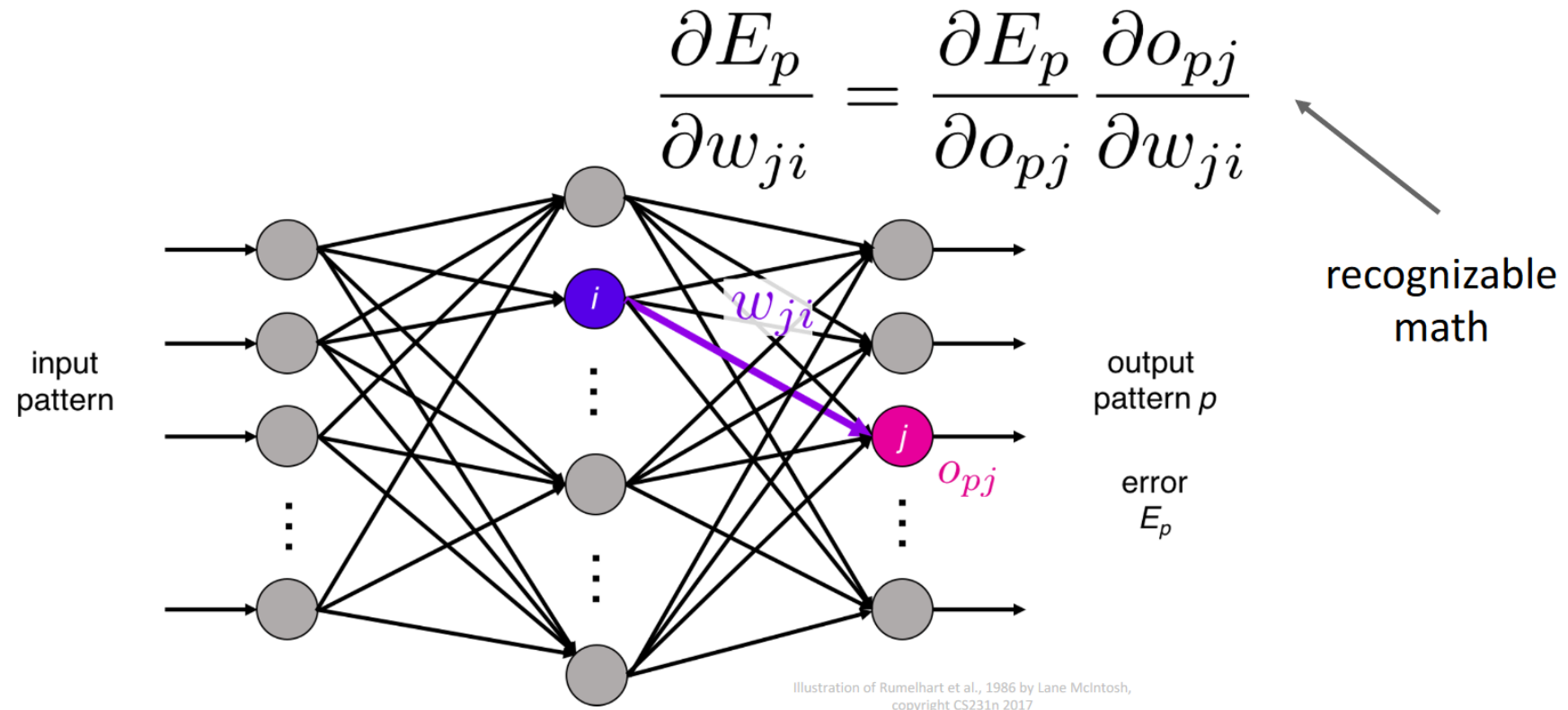
No practical training algorithm



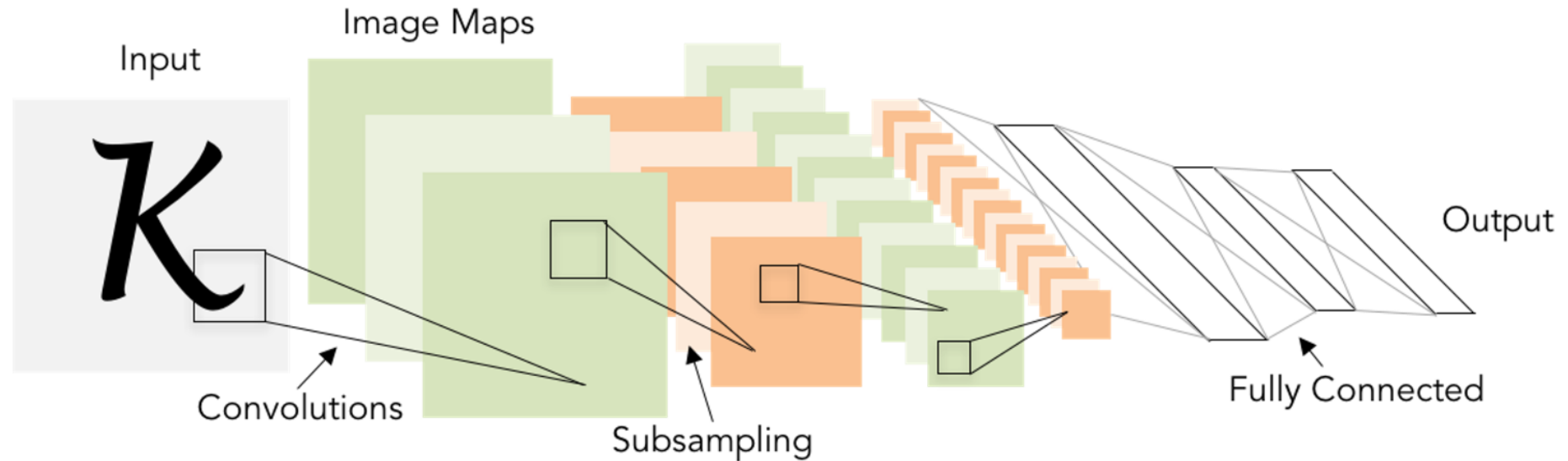
Backprop: Rumelhart, Hinton, and Williams, 1986

Introduced
backpropagation for
computing gradients
in neural networks

Successfully trained
perceptrons with
multiple layers



Convolutional Networks: LeCun et al, 1998



Applied backprop algorithm to a Neocognitron-like architecture

Learned to recognize handwritten digits

Was deployed in a commercial system by NEC, processed handwritten checks

Very similar to our modern convolutional networks!

2000s: “Deep Learning”

People tried to train neural networks that were deeper and deeper

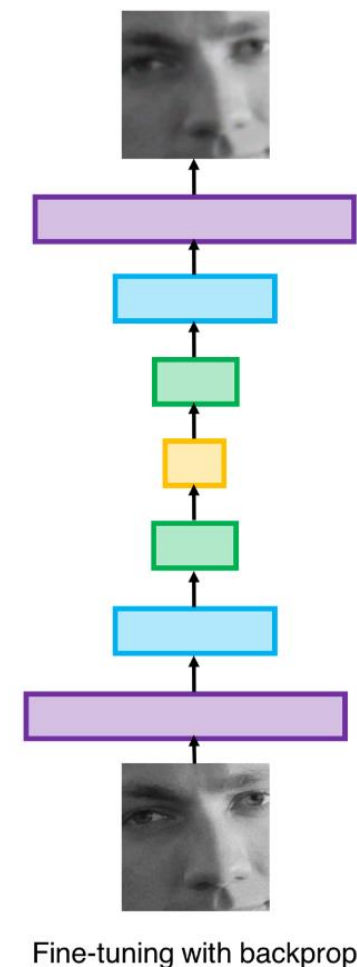
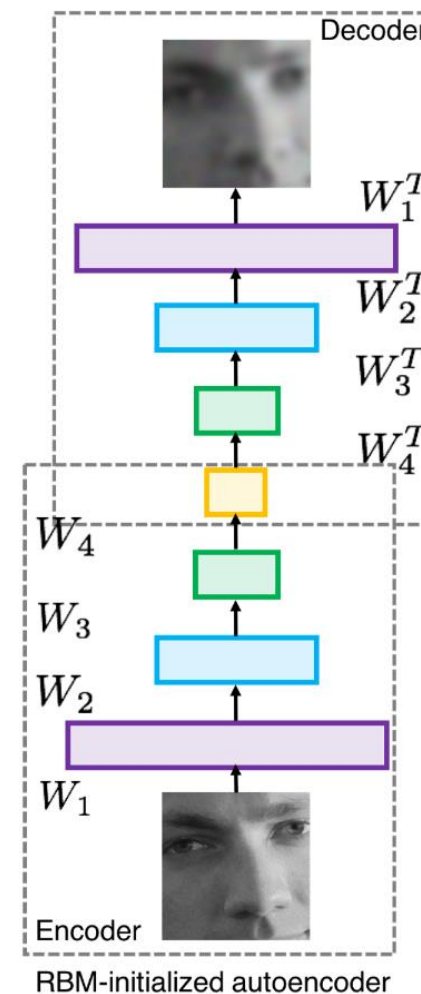
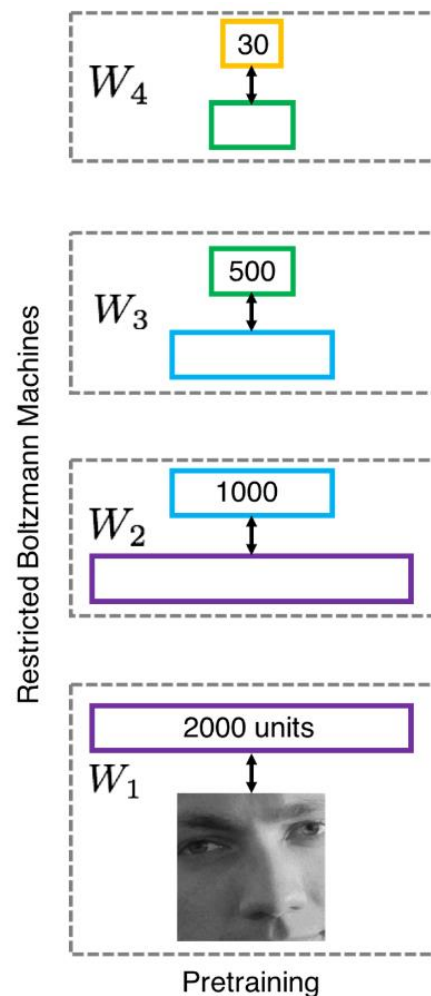
Not a mainstream research topic at this time

Hinton and Salakhutdinov, 2006

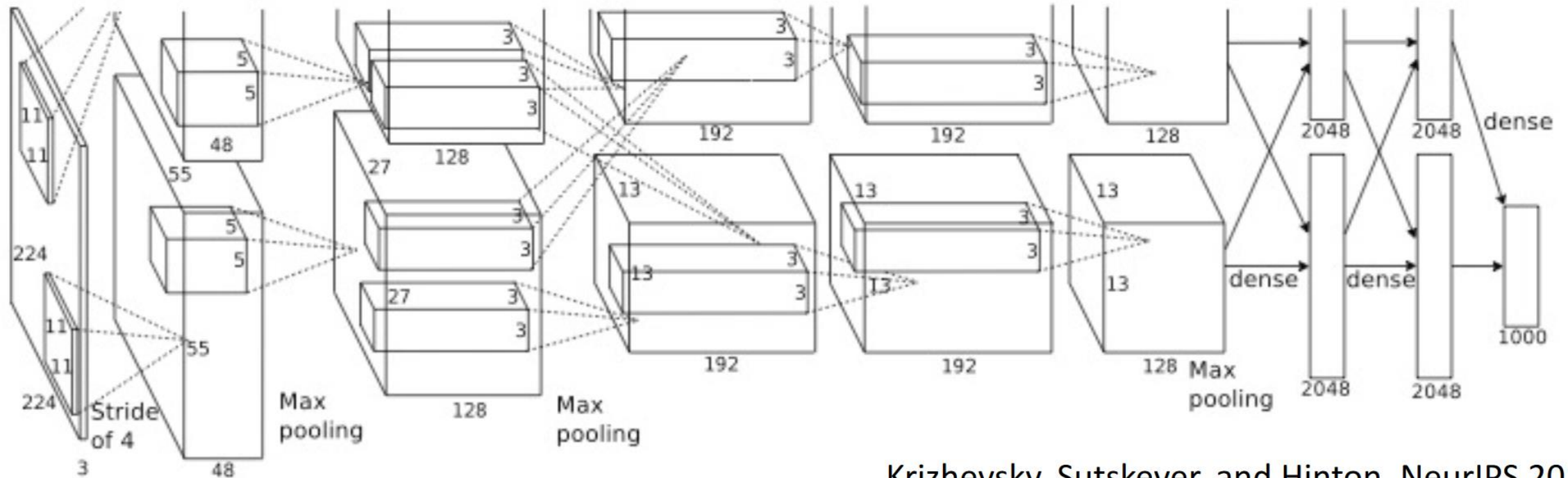
Bengio et al, 2007

Lee et al, 2009

Glorot and Bengio, 2010



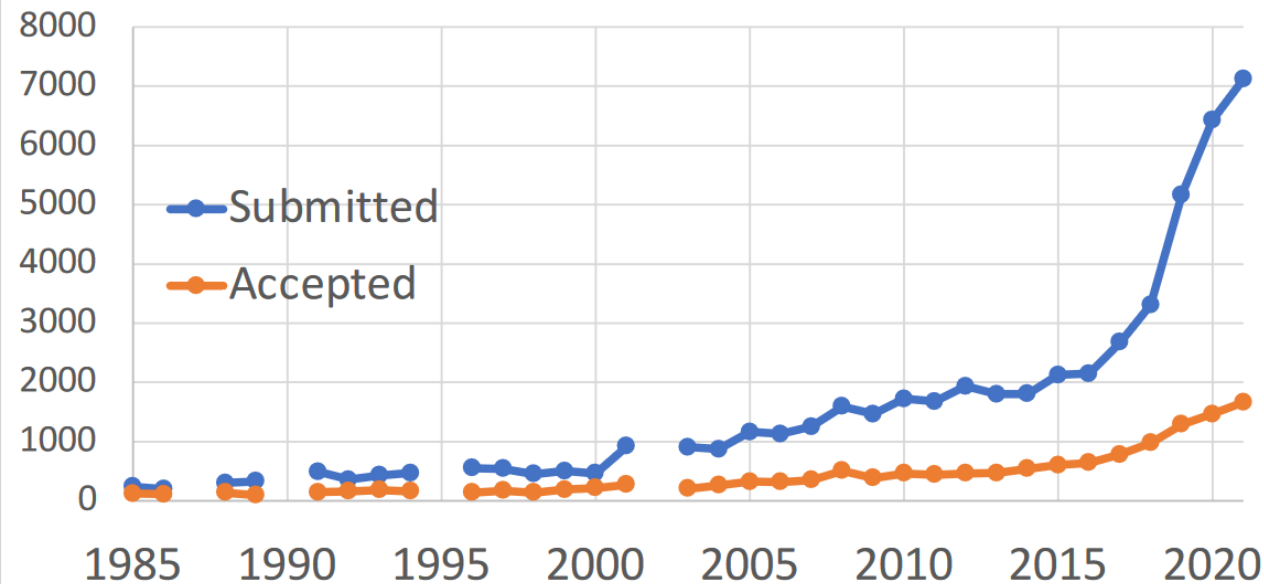
AlexNet: Deep Learning Goes Mainstream



Krizhevsky, Sutskever, and Hinton, NeurIPS 2012

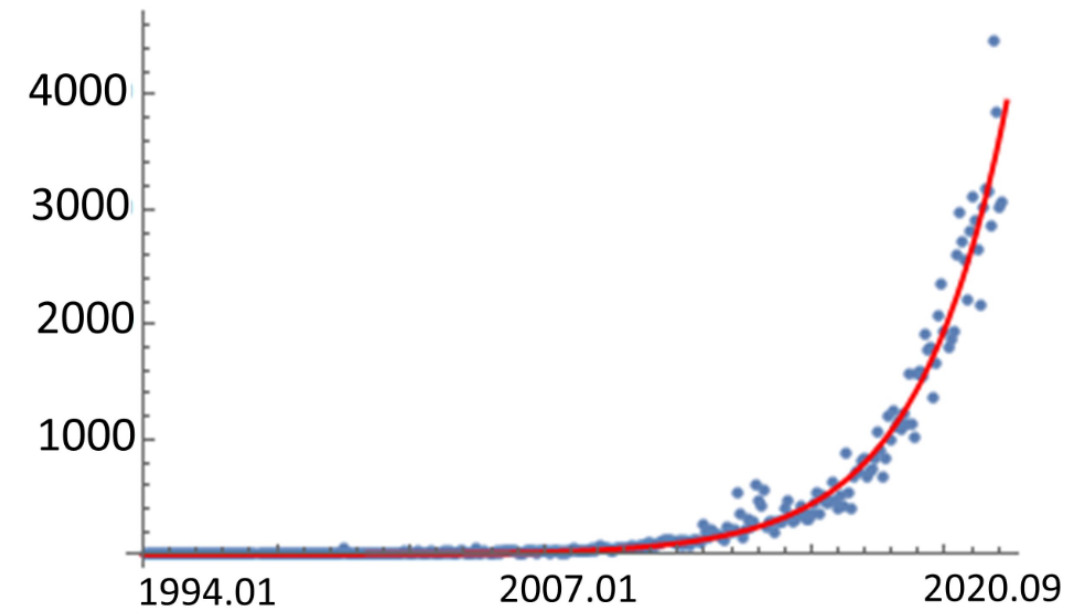
2012 to Present: Deep Learning Explosion

CVPR Papers

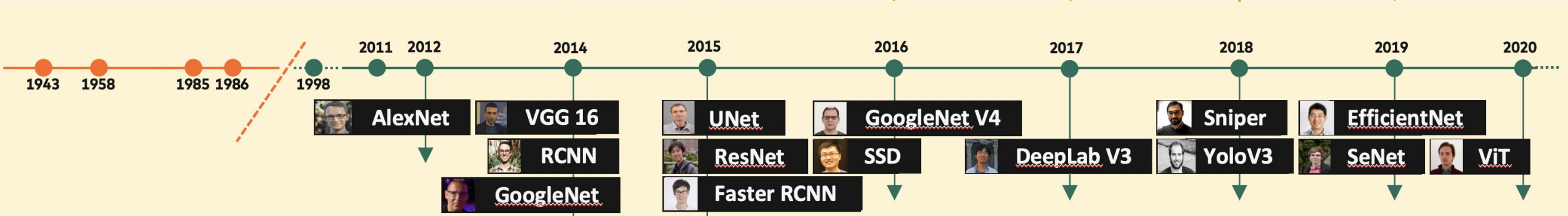
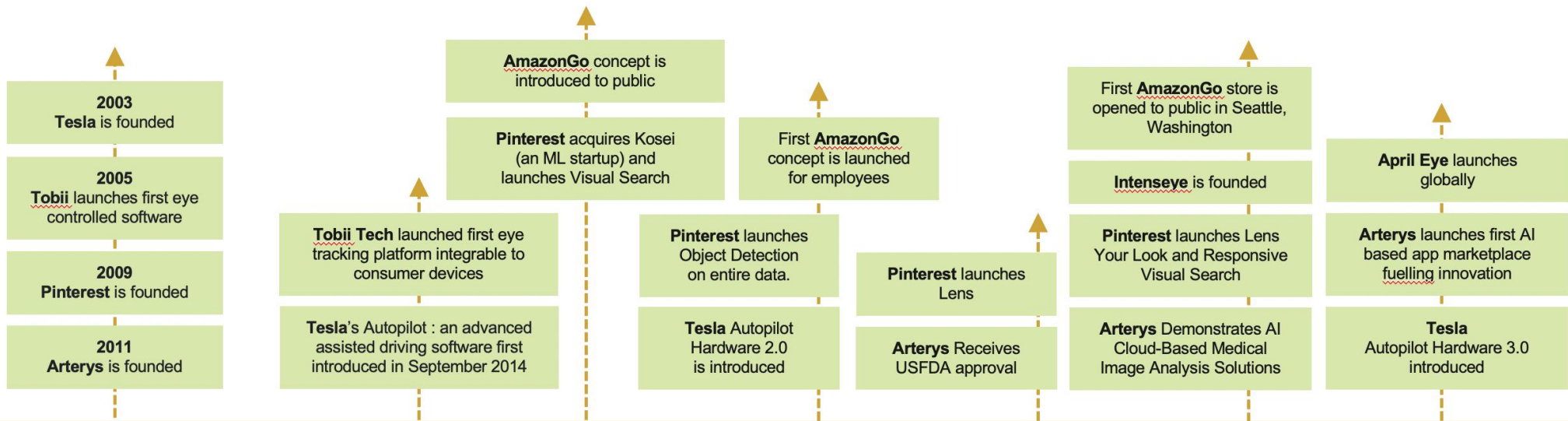
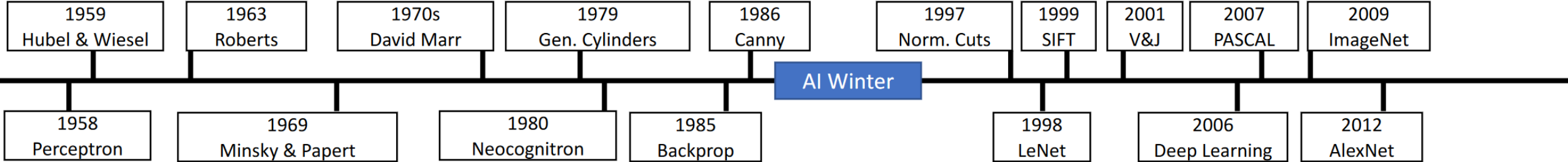


Publications at top Computer Vision conference

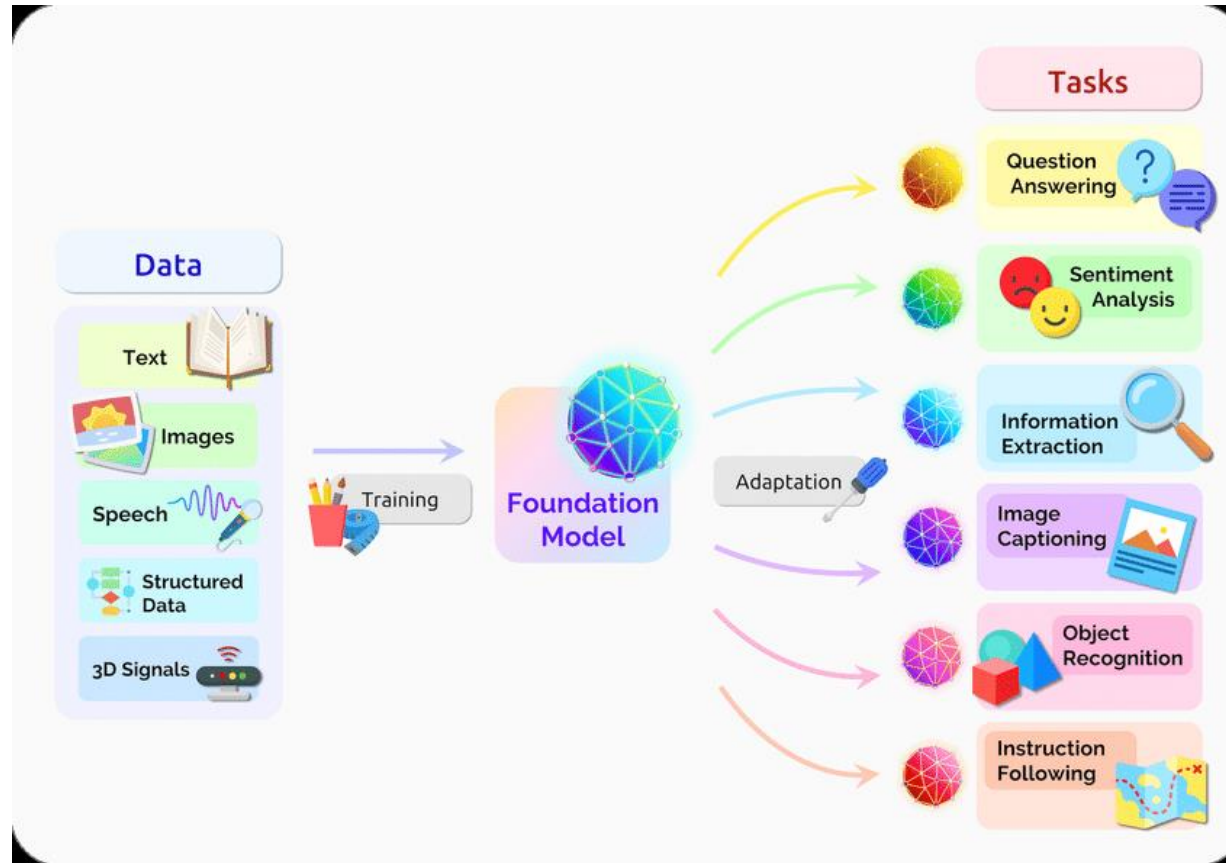
ML+AI arXiv papers per month



arXiv papers per month ([source](#))



Multi-modal foundation models – current trends



Can we learn a common and generalized model from multiple modalities of data, that can solve different downstream tasks all at once?

Resources

- Online lectures (**Michigan**, **NYU**, **TUM**)
- <https://d2l.ai/d2l-en.pdf>
- <https://www.cs.princeton.edu/courses/archive/fall19/cos597B/lecnotes/bookdraft.pdf>
- <https://github.com/janishar/mit-deep-learning-book-pdf>
- <https://tanthiamhuat.files.wordpress.com/2018/03/deeplearningwithpython.pdf>

Pytorch, Tensorflow, Keras

- Pytorch – zero to GAN
 - <https://www.youtube.com/watch?v=Glsg-ZUy0MY>
- Keras by deepLizard
 - <https://www.youtube.com/watch?v=qFJeN9V1Zsl>
- Tensorflow tutorial
 - <https://www.youtube.com/watch?v=tPYj3fFJGjk>